

Reconstruction of Function Fields from Mod- ℓ Milnor K -Theory

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Abstract

We prove a mod- ℓ Bogomolov–Pop reconstruction theorem for function fields of transcendence degree at least two over arbitrary algebraically closed fields of characteristic different from ℓ . The groups K_1^M/ℓ and K_2^M/ℓ , together with their full bilinear product, determine the perfect closure and its constant field. Every compatible isomorphism of these data is induced by a field isomorphism up to a single scalar in $\mathbb{F}_\ell^\times$, with only Frobenius ambiguity in the field isomorphism in positive characteristic.

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1 Introduction

Birational anabelian geometry asks how much of a function field can be recovered from its Galois-theoretic invariants. In the Bogomolov program, the field has algebraically closed constants, and the relevant invariants are much smaller than the full absolute Galois group. The present paper proves the reconstruction assertion for the first two Milnor K -groups modulo a single prime, with their multiplication. The result includes the isomorphism statement: it identifies every compatible linear isomorphism, as well as the ambiguity in the field isomorphism that induces it.

1.1 The reconstruction theorem

Fix a prime ℓ and put $\Lambda = \mathbb{F}_\ell$. For a field F of characteristic different from ℓ , set

$$V_F = F^\times / (F^\times)^\ell, \quad R_F = \text{span}_\Lambda \{[x] \otimes [1-x] : x \in F \setminus \{0, 1\}\}.$$

We write V_F additively, so $[xy] = [x] + [y]$. Let

$$W_F = (V_F \otimes_\Lambda V_F) / R_F, \quad m_F : V_F \otimes_\Lambda V_F \longrightarrow W_F.$$

Thus $V_F = K_1^M(F)/\ell$ and $W_F = K_2^M(F)/\ell$, with m_F the Milnor product. A Λ -linear isomorphism $\Theta : V_K \rightarrow V_L$ is *compatible* if

$$(\Theta \otimes \Theta)(R_K) = R_L.$$

Equivalently, there is a unique linear isomorphism $\Theta_2 : W_K \rightarrow W_L$ intertwining the products. Write $\text{Isom}_M(V_K, V_L)$ for the set of compatible isomorphisms. Multiplication by $a \in \Lambda^\times$ acts on this set; the induced degree-two map is then multiplied by a^2 .

Let K/k and L/l be finitely generated extensions of algebraically closed fields. Write F^i for the perfect closure of a field F : it equals F in characteristic zero and $\bigcup_{r \geq 0} F^{1/p^r}$ in characteristic $p > 0$. Denote by $\text{Isom}^i(K, L)$ the field isomorphisms $\alpha : K^i \rightarrow L^i$ satisfying $\alpha(k) = l$. In positive characteristic, identify two such isomorphisms if they differ by postcomposition with Frob_L^n , $n \in \mathbb{Z}$, where $\text{Frob}_L(z) = z^p$. Denote the resulting set by $\text{Isom}_F^i(K, L)$; in characteristic zero take no quotient, and if the characteristics differ this set is empty.

Purely inseparable extension induces canonical isomorphisms on V and W . Indeed, for an extension of exponent at most r , inclusion and the p^r -power homomorphism compose to the p^r -power map on each field. On Milnor groups in degrees one and two this composition is multiplication by p^r and p^{2r} , respectively, both invertible modulo ℓ . Passing to the union proves the assertion for perfect closures. Thus α induces a compatible map $\alpha_1 : V_K \rightarrow V_L$, and postcomposition by Frob_L^n multiplies it by p^n in $\Lambda^\times$. There is consequently a canonical map

$$\text{Isom}_F^i(K, L) \longrightarrow \text{Isom}_M(V_K, V_L) / \Lambda^\times. \quad (1.1)$$

Theorem 1.1. *Let ℓ be any prime. Let K/k and L/l be finitely generated extensions of arbitrary algebraically closed fields, with $\text{char } k, \text{char } l \neq \ell$ and $\text{trdeg}(K/k), \text{trdeg}(L/l) \geq 2$. Then (1.1) is bijective. In particular, a compatible isomorphism $V_K \rightarrow V_L$ forces equality of the characteristics and of the relative transcendence degrees.*

Only the two vector spaces and their bilinear product enter this statement. The proof recovers the valuations and rational-subfield images needed for reconstruction from this datum. The scalar is one global element of $\mathbb{F}_\ell^\times$; at $\ell = 2$ this ambiguity is trivial.

1.2 Historical context

Bogomolov proposed reconstructing higher-dimensional function fields over algebraically closed constants from the pro- ℓ quotient in which commutators are central [1]. Bogomolov and Tschinkel proved reconstruction for surfaces over algebraic closures of finite fields [2], and subsequently for higher-dimensional function fields over those constants [4]. Pop independently completed reconstruction in this constant-field setting [7]. These results established that a small part of the Galois group can retain the geometry needed to recover a field. Over more general algebraically closed constants, Pop also obtained reconstruction from pro- ℓ data endowed with divisorial inertia, in relative transcendence degree greater than two [9, Theorem 1.1].

The finite-coefficient problem requires additional control of the global reconstruction. Topaz proved a Milnor-theoretic isomorphism theorem in relative transcendence degree at least five when the degree-one and degree-two mod- ℓ groups and their product are supplied together with all rational subgroups [12, Theorem B]. Those subgroups are the images of the multiplicative groups of relatively algebraically closed rational one-variable subfields. Theorem 1.1 obtains the required rational subfields from the product itself, and its point-value argument applies in every relative dimension at least two.

Algebraic dependence has provided another route from multiplicative invariants to field structure. Bogomolov and Tschinkel used Milnor K -theory modulo infinitely divisible elements in characteristic zero [3]. Cadoret and Pirutka reconstructed regular function fields over perfect constants from the multiplicative quotient by constants together with algebraic dependence, and derived applications to integral Milnor K -theory [5]. Topaz later proved reconstruction from rational Milnor K -theory in absolute transcendence degree at least five [15]. These results clarify the role of algebraic dependence, while using invariants different from the single-prime datum considered here.

Our local inputs are the alternating-pair valuation theory recorded in [13, Section 2] and Pop’s density theorem for minimized inertia [8]. We state the precise forms used below. The bounded-support and incidence method is adapted from [6, Sections 4–6]. The finite-coefficient bounded-support and incidence arguments are proved here. The support estimate also has an antecedent in the bounded-genus curve sections and Hurwitz argument of Bogomolov and Tschinkel [4, Section 6, Proposition 6.1]. The proof uses no pro- ℓ field-reconstruction theorem as a premise.

1.3 The proof and its main ingredients

The argument begins with a compatible map Θ and its dual on the compact character spaces $A_F = \text{Hom}(F^\times, \Lambda)$. Two characters f, g form an *alternating pair* when $f(x)g(1-x) = f(1-x)g(x)$ for every $x \neq 0, 1$. This condition is visible in the Milnor product and is therefore preserved by Θ . Established local theory turns alternating subspaces into valuations. Section 2 uses it to recover the transcendence degree and the inertia and decomposition spaces of quasi-divisorial valuations, and the character spaces of their iterated residue fields. These valuations may act nontrivially on the constants; their residue constants can therefore have positive characteristic even when the original field has characteristic zero.

The next task is to recover curve subfields: relatively algebraically closed intermediate fields $k \subset E \subset K$ of transcendence degree one over k . Individual mod- ℓ classes record only orders modulo ℓ , so a divisor can disappear when its multiplicity is divisible by ℓ . We control this loss by passing to residual curves. Section 3 uses Pop’s density theorem to recognize residual curves whose constants are algebraic closures of finite fields, together with all their point order lines.

Section 4 then constructs a residual curve of bounded genus for any finite independent list of classes in V_L . It preserves independence and realizes each support degree on a fixed projective model of L as the number of nonzero point orders on the residual curve. For each nonconstant $t \in K$, applying Riemann–Hurwitz to five fixed members of its pencil bounds the total degree of the divisor support of every class $\Theta([t - a])$, $a \in k$, on that model.

For a curve subfield E , choose $t \in E \setminus k$. The supporting prime divisors of the classes $\Theta([t - a])$ form an infinite set of bounded degree, hence lie in a finite-type family. The valuation correspondence shows that each class in $\Theta(V_E)$ has an ℓ -th-power residue along all but finitely many of these divisors. Section 5 studies the incidence variety of pairs consisting of a point and a divisor containing it. Restricting its parameter space to a curve produces a finite extension of L containing the curve’s function field. After passage to this extension, the classes come from that one-variable parameter field. The argument then descends them to a curve subfield $P \subset L$ and proves $\Theta(V_E) = V_P$.

Two features are useful specifically for finite coefficients. One parameter curve works for every class. For each class, a finite change of parameter space allows normality to extend a generic ℓ -th root of a representative across the smooth locus where that representative is a unit. This avoids any restriction on the cardinality of the constants. Moreover, the incidence field is the compositum of L and the parameter field. This identity permits exact descent, retaining information that a norm could annihilate modulo ℓ .

Once curve subfields correspond, Section 6 recognizes true divisors, which are trivial on the constants, and a sufficiently large family of rational subfields on which every point is detected. A quotient of two local parameters at a smooth point supplies such a subfield. These quotients generate the function field, and the construction works already in dimension two. Matching their point-order lines gives bijections between the corresponding projective lines of points.

Section 7 recovers field operations from simultaneous values of these rational functions. A blowup at a smooth point detects all coordinates of a finite tuple at once. Additive triples yield rational expressions, allowing p -power roots in characteristic p , for the transported addition law. Their one-variable slices align all point bijections, outside finite sets, with a single isomorphism of constant fields. An identity of rational functions removes the apparent freedom caused by finite exceptional sets in those slices. The resulting correspondence preserves every algebraic relation among the chosen generators and hence gives an isomorphism of perfect fields.

Finally, Section 8 proves that this isomorphism induces the original map up to one scalar, and that scalar action of a field automorphism forces it to be a Frobenius power. Throughout, the pro- ℓ density input concerns individual fields. The given mod- ℓ isomorphism is never lifted to a pro- ℓ isomorphism.

1.4 Conventions

A function field F/κ means a finitely generated field extension. Its constants κ will be algebraically closed. A curve is a geometrically integral variety of dimension one, and its function field has a unique smooth projective model over algebraically closed constants. A field extension is regular if it is separable and its base is relatively algebraically closed. Points of varieties over an algebraically closed field mean rational points unless a scheme point is specified. The phrase “for general points” means on some dense Zariski open subset. We continue to use $V_F = F^\times / (F^\times)^\ell$ for residue fields of characteristic ℓ , as a multiplicative quotient, without a Galois-cohomological interpretation.

2 Alternating characters and quasi-divisorial valuations

We first recover valuations from the degree-two relations. The local theory is most naturally expressed on the dual of the multiplicative group. Its multiplicative formulation also applies to residue fields of characteristic ℓ , which will occur in the argument.

For any field F , put

$$A_F = \text{Hom}(F^\times, \Lambda) = \text{Hom}_\Lambda(V_F, \Lambda), \quad \Lambda = \mathbb{F}_\ell.$$

Here the definition of V_F is used in every characteristic. Give A_F the topology of pointwise convergence, with Λ discrete. Choosing a basis of V_F identifies A_F with a product of copies of Λ . In particular it is compact, and evaluation identifies V_F with $\text{Hom}_{\text{cont}}(A_F, \Lambda)$: a continuous linear form on that product depends on only finitely many coordinates.

A pair $f, g \in A_F$ is *alternating* if

$$f(x)g(1-x) = f(1-x)g(x) \quad (x \in F \setminus \{0, 1\}). \quad (2.1)$$

A subset is alternating if every pair of its elements is alternating. The determinant functional

$$[x] \otimes [y] \mapsto f(x)g(y) - f(y)g(x)$$

annihilates R_F exactly when f, g are alternating. Thus the isomorphism Θ in Theorem 1.1 induces a continuous linear isomorphism

$$\Phi = \Theta^* : A_L \xrightarrow{\sim} A_K$$

that preserves alternating pairs in both directions.

We consider valuations up to equivalence and write $v \leq w$ when v is a coarsening of w . Write U_v and U_v^1 for its units and principal units, Fv for its residue field, and vF for its value group. The *minimized inertia and decomposition spaces* are

$$I_v = U_v^\perp \subset D_v = (U_v^1)^\perp \subset A_F,$$

where perpendiculars refer to evaluation. A subset of A_F is *valuative* if it is contained in some I_v . Restriction to units gives a continuous map

$$\rho_v : D_v \longrightarrow A_{Fv}$$

with kernel I_v . Surjectivity will be established for the valuations we use below.

2.1 The established local input

The following form of alternating-pair valuation theory collects [13, Fact 2.1, Lemma 2.5, Theorem 2.6, and Corollary 2.7]. We state it for finite coefficients; the field in this statement need not have characteristic different from ℓ .

Theorem 2.1 (Alternating-pair valuation theory). *Let F be a field and set $A_F^\pm = \{f \in A_F : f(-1) = 0\}$.*

- (i) *Every valuative subset $\Sigma \subset A_F$ has a unique coarsest valuation v_Σ with $\Sigma \subset I_{v_\Sigma}$. In particular, v_Σ coarsens every valuation whose inertia contains Σ . For such a valuation w , this coarsening is obtained by quotienting wF by its largest convex subgroup contained in $w(\bigcap_{f \in \Sigma} \ker f)$.*

- (ii) If $S \subset A_F^\pm$ is an alternating linear subspace, its valutive elements form a valutive subspace S_0 of codimension at most one. Moreover $S \subset D_{v_{S_0}}$.
- (iii) If $\Sigma \subset A_F^\pm$ is valutive and $g \in A_F^\pm$ is alternating with every element of Σ , then $g \in D_{v_\Sigma}$.

The coefficient convention in [13, Section 2.1] explicitly allows $\mathbb{Z}/\ell$; its fraction field is then $\mathbb{F}_\ell$, so its character space is exactly A_F . The case $\ell = 2$ is included in the local theorem. For the fields over algebraically closed constants considered here, every character kills the constants and hence -1 , so $A_F^\pm = A_F$.

2.2 Value groups and residue characters

For the remainder of this section, let F/κ be a finitely generated extension of an algebraically closed field, and put $s = \text{trdeg}(F/\kappa)$. No restriction is imposed on its characteristic. We first record the valuation facts that turn the local theorem into an intrinsic description of codimension-one valuations.

Lemma 2.2. *For every valuation v of F , the field κv is algebraically closed, the group $vF/v\kappa$ is torsion free, and*

$$\text{rrank}(vF/v\kappa) + \text{trdeg}(Fv/\kappa v) \leq s. \quad (2.2)$$

Here $\text{rrank}(G) = \dim_{\mathbb{Q}}(G \otimes_{\mathbb{Z}} \mathbb{Q})$. If equality holds, then $Fv/\kappa v$ is finitely generated and $vF/v\kappa$ is a finitely generated free abelian group. Equality passes to every coarsening and to the induced residue valuation. Conversely, equality for a valuation and an induced residue valuation gives equality for their composition.

Proof. The value group of an algebraically closed field is divisible, and its residue field is algebraically closed. If $n\gamma \in v\kappa$ for $\gamma \in vF$, choose $\delta \in v\kappa$ with $n\delta = n\gamma$. Ordered abelian groups are torsion free, so $\gamma = \delta$. This proves torsion-freeness of the relative group.

Choose elements $x_1, \dots, x_a \in F^\times$ whose values are rationally independent modulo $v\kappa$, and units $y_1, \dots, y_b$ whose residues are algebraically independent over κv . These elements are algebraically independent over κ . Indeed, in a putative polynomial relation, distinct monomials in the x_i have distinct values modulo $v\kappa$. Within a coefficient polynomial in the y_j , its terms of least value cannot cancel, by algebraic independence of their residues after rescaling by a constant. The nonzero summands obtained by grouping according to monomials in the x_i have distinct values, so their sum cannot vanish. This proves (2.2) by taking maximal such families.

Suppose equality holds and choose these families with $a + b = s$. Then F is finite over $F' = \kappa(x_1, \dots, x_a, y_1, \dots, y_b)$. The same least-value calculation gives

$$vF'/v\kappa \cong \mathbb{Z}^a, \quad F'v = \kappa v(\bar{y}_1, \dots, \bar{y}_b).$$

The fundamental inequality for the finite extension F/F' shows that $[vF : vF']$ and $[Fv : F'v]$ are finite. Thus the residue extension is finitely generated, while the relative value group is finitely generated and torsion free, hence free.

For the assertion about composition, write $v = w \circ u$, meaning that u is a coarsening and w is the induced valuation on Fu . The value groups fit into the exact sequence

$$0 \longrightarrow w(Fu)/w(\kappa u) \longrightarrow vF/v\kappa \longrightarrow uF/u\kappa \longrightarrow 0.$$

Apply (2.2) first to u , and then to w on $Fu/\kappa u$. The sum of the two inequalities is precisely the inequality for v . Consequently equality for v forces equality in both, and equality in both forces

equality for v . This use of the inequality on $Fv/\kappa v$ is legitimate even before finite generation is known: the algebraic-independence argument proving it applies to arbitrary field extensions of finite transcendence degree. $\square$

Definition 2.3. A *quasi-divisorial valuation*, or *quasi-prime divisor*, of F/κ is a valuation minimal under coarsening among those with

$$vF/v\kappa \cong \mathbb{Z}, \quad \text{trdeg}(Fv/\kappa v) = s - 1.$$

A quasi-prime divisor trivial on κ is called a *prime divisor*, or a *true prime divisor*. A *quasi-prime r -divisor* is a composition of r successive quasi-prime divisors on the resulting residue fields.

Lemma 2.2 shows that for a quasi-prime r -divisor v , the residue field is a function field of transcendence degree $s - r$ over κv , and

$$vF/v\kappa \cong \mathbb{Z}^r. \quad (2.3)$$

The isomorphism concerns abstract abelian groups; no ordering of $\mathbb{Z}^r$ is specified.

Lemma 2.4. Suppose $vF/v\kappa$ is free abelian of finite rank. Then restriction to units induces a canonical topological isomorphism

$$D_v/I_v \xrightarrow{\sim} A_{Fv}. \quad (2.4)$$

Two elements of D_v are alternating if and only if their images in A_{Fv} are alternating. If w is a valuation of Fv , then

$$I_{w \circ v} = \rho_v^{-1}(I_w), \quad D_{w \circ v} = \rho_v^{-1}(D_w), \quad (2.5)$$

where the inverse images are taken inside D_v .

Proof. There is an exact sequence of abelian groups

$$1 \longrightarrow (Fv)^\times / (\kappa v)^\times \longrightarrow F^\times / (\kappa^\times U_v^1) \longrightarrow vF/v\kappa \longrightarrow 0.$$

For the kernel assertion, rescale by a constant an element whose value lies in $v\kappa$, and then take its residue. The last group is free, so the sequence splits as a sequence of abstract groups. Dualizing into Λ is consequently exact. Characters kill the algebraically closed constants on both sides, giving the asserted surjection ρ_v with kernel I_v . Its induced bijection is a homeomorphism because its source is compact and its target Hausdorff.

For alternation, if $v(x) > 0$, then $1 - x \in U_v^1$, and the identity (2.1) holds for any pair in D_v . The case $v(1 - x) > 0$ is the same. If $v(x) < 0$, then $(1 - x)/(-x) \in U_v^1$, so every character in D_v takes the same value on x and $1 - x$. The remaining case is $v(x) = v(1 - x) = 0$, when the identity is exactly its residue-field counterpart. Conversely, lift any residue element different from zero and one to test the residue identity.

Finally U_v^1 is contained in both $U_{w \circ v}$ and $U_{w \circ v}^1$. Under reduction of v -units, these two groups map respectively onto U_w and U_w^1 . Their annihilators therefore give (2.5). $\square$

We will also use the following consequence of valuation approximation. It explains why intersections of decomposition spaces yield inertia.

Lemma 2.5. Let v_1, v_2 be incomparable valuations of a field, and let u be their finest common coarsening. Then

$$U_{v_1}^1 U_{v_2}^1 = U_u, \quad D_{v_1} \cap D_{v_2} = I_u.$$

Proof. The induced valuations w_1, w_2 on the residue field Fu are independent. For $a \in (Fu)^\times$, the approximation theorem gives $b \in (Fu)^\times$ with

$$w_1(b - 1) > 0, \quad w_2(b - a) > w_2(a).$$

Thus $b \in U_{w_1}^1$ and $a/b \in U_{w_2}^1$, proving $U_{w_1}^1 U_{w_2}^1 = (Fu)^\times$. Reduction maps $U_{v_i}^1$ onto $U_{w_i}^1$, and both groups $U_{v_i}^1$ contain U_u^1 . Lifting the product identity gives the first equality; taking annihilators gives the second. $\square$

2.3 Recovering dimension and quasi-divisors

The next argument follows the alternating-space method of [13, Fact 3.2 and Theorem 3.3]. We give the details so that the argument remains available in characteristic ℓ .

Proposition 2.6. *The maximum dimension of an alternating subspace of A_F is s . If $S \subset A_F$ is alternating of dimension s , and v is the valuation associated to its valutive part S_0 , then $S_0 = I_v$ and equality holds in (2.2) for v .*

Proof. For any torsion-free abelian group G of finite rational rank,

$$\dim_\Lambda(G/\ell G) \leq \text{rrank}(G).$$

Indeed, representatives of linearly independent classes modulo ℓ are rationally independent: an integral relation can be divided by the greatest common divisor of its coefficients, using torsion-freeness, and the resulting relation has a coefficient nonzero modulo ℓ . Since characters of vF kill its divisible subgroup $v\kappa$, this proves

$$\dim I_v \leq \text{rrank}(vF/v\kappa).$$

Now let S be alternating. By Theorem 2.1, its valutive part S_0 is a subspace of codimension at most one and, for its associated valuation v , $S_0 \subset I_v$ and $S \subset D_v$. Moreover $S \cap I_v = S_0$, since every element of I_v is valutive. If $S = S_0$, then

$$\dim S \leq \dim I_v \leq \text{rrank}(vF/v\kappa) \leq s.$$

If $S \neq S_0$, an element of $S \setminus S_0$ has nonzero image under ρ_v , whose kernel is always I_v . Thus $A_{Fv} \neq 0$. Since κv is algebraically closed, this implies $\text{trdeg}(Fv/\kappa v) \geq 1$, and hence

$$\dim S = \dim S_0 + 1 \leq \text{rrank}(vF/v\kappa) + \text{trdeg}(Fv/\kappa v) \leq s.$$

If $\dim S = s$, equality throughout the applicable chain gives $S_0 = I_v$ and equality in (2.2).

To attain the bound, choose a transcendence basis of F/κ , take a full coordinate flag on its rational function field, and prolong the associated valuation to F . It is trivial on κ , and its value group is a finite-index extension of $\mathbb{Z}^s$, hence is abstractly $\mathbb{Z}^s$. Its inertia has dimension s and is alternating by Lemma 2.4. For $s = 0$, the field is κ , its character space is zero, and the assertion has the same interpretation. $\square$

We can now identify the one-dimensional inertia spaces solely through maximal alternating subspaces. Once inertia is identified, alternation with it identifies decomposition as well.

Proposition 2.7. *Assume $s \geq 2$. A line $H \subset A_F$ is the inertia of a quasi-divisorial valuation if and only if*

$$H = S_1 \cap S_2 \quad \text{for alternating subspaces } S_1, S_2 \subset A_F \text{ of dimension } s. \quad (2.6)$$

The quasi-divisorial valuation v with $I_v = H$ is unique. For any nonzero $h \in H$, its decomposition space is

$$D_v = \{g \in A_F : (h, g) \text{ is alternating}\}. \quad (2.7)$$

Proof. Suppose first that v is quasi-divisorial. On $Fv/\kappa v$ choose two independent full discrete flag valuations w_1, w_2 , trivial on κv . To obtain them, start with coordinate flags on a rational subfield of transcendence degree $s - 1$, with distinct first prime divisors, and prolong to the finite extension Fv . Independence survives prolongation. Indeed, a valuation on a finite field extension that restricts trivially to the smaller field is trivial, since its value group is then finite and hence zero. Thus a common nontrivial coarsening of w_1, w_2 would restrict to a common nontrivial coarsening of the original independent flag valuations, which is impossible.

The inertia spaces I_{w_1}, I_{w_2} have zero intersection by approximation. Set $v_i = w_i \circ v$ and $S_i = I_{v_i}$. Each v_i has relative value group $\mathbb{Z}^s$, so its inertia is alternating of dimension s . By (2.5), the intersection $S_1 \cap S_2$ is I_v .

Conversely, suppose (2.6) holds. Let $S_{i,0}$ be the valuative part of S_i , with associated valuation v_i . Proposition 2.6 gives

$$S_{i,0} = I_{v_i}, \quad S_i \subset D_{v_i}, \quad \dim I_{v_i} \geq s - 1 \geq 1,$$

and v_i has equality in (2.2). We first show that H is valuative. If, for example, $v_1 \leq v_2$, then

$$0 \neq I_{v_1} \subset S_1 \cap S_2 = H,$$

so $H = I_{v_1}$. The other comparable case is symmetric. If the valuations are incomparable, Lemma 2.5 puts H inside the inertia of their finest common coarsening. In either case every element of H is valuative. Because it lies in each S_i , it then lies in each $S_{i,0}$, giving

$$H = I_{v_1} \cap I_{v_2}.$$

Let $u = v_H$. By Theorem 2.1, it coarsens each v_i , and therefore

$$H \subset I_u \subset I_{v_1} \cap I_{v_2} = H.$$

Equality in (2.2) passes to u . Its relative value group is free, so $\dim I_u = 1$ implies $uF/u\kappa \cong \mathbb{Z}$ and $\text{trdeg}(Fu/\kappa u) = s - 1$. If a coarsening $u' \leq u$ still has relative rank one, its one-dimensional inertia is contained in I_u , hence equals H . The defining coarseness of $v_H = u$ then gives $u \leq u'$, so $u = u'$. Thus u is quasi-divisorial.

For uniqueness, if v is any quasi-divisor with $I_v = H$, its associated coarsening v_H has the same inertia. Equality in (2.2) passes to that coarsening, so it too has relative rank one and residue transcendence degree $s - 1$. Minimality of v forces $v = v_H$.

Finally, an element of D_v is alternating with h by Lemma 2.4, since h has zero residue image. Conversely, an element alternating with h belongs to $D_{v_H} = D_v$ by Theorem 2.1. $\square$

Lemma 2.8. *If v is a quasi-prime r -divisor, with $r \geq 1$, then its value group has no nonzero ℓ -divisible convex subgroup. Moreover $v = v_{I_v}$; in particular its inertia determines the valuation.*

Proof. First suppose $r = 1$. An ℓ -divisible convex subgroup C of vF maps to zero in $vF/v\kappa \cong \mathbb{Z}$, and hence lies in $v\kappa$. Coarsening by C therefore leaves the relative value group unchanged. Lemma 2.2 shows that the residue transcendence degree remains $s - 1$, so minimality of v forces $C = 0$.

For a composite flag, the value group of its final quasi-prime step is a nonzero convex subgroup $B \subset vF$ with the property just proved. If $C \subset vF$ were a nonzero ℓ -divisible convex subgroup, then $B \cap C \neq 0$, since convex subgroups are linearly ordered by inclusion. This intersection is ℓ -divisible: division by ℓ in C stays in B by convexity. It is also convex in B , a contradiction.

Put $\Gamma = vF$. Since $\Gamma/v\kappa \cong \mathbb{Z}^r$ and $v\kappa$ is divisible, the common kernel in Γ of all characters in I_v is

$$v\kappa + \ell\Gamma = \ell\Gamma.$$

Every convex subgroup contained in $\ell\Gamma$ is itself ℓ -divisible: an ℓ -th part exists in Γ , and convexity places it in the subgroup. The first assertion therefore shows that the largest such convex subgroup is zero. The coarsening description in Theorem 2.1 now gives $v = v_{I_v}$. $\square$

Corollary 2.9. *For the fields $K/k, L/l$ of Theorem 1.1, a compatible isomorphism Θ forces $\text{trdeg}(K/k) = \text{trdeg}(L/l) = d$, say. Its dual Φ matches the inertia–decomposition pairs of quasi-prime divisors, and, successively, the pairs of quasi-prime r -divisors for $1 \leq r \leq d - 1$. For a matched pair of valuations v on K and w on L , it induces a topological linear isomorphism*

$$A_{Lw} \xrightarrow{\sim} A_{Kv}$$

preserving alternating pairs in both directions.

Proof. The maximum alternating dimension is preserved by Φ , giving equality of transcendence degrees by Proposition 2.6. Proposition 2.7 then identifies corresponding quasi-prime inertia and decomposition spaces. Quotienting by inertia and using Lemma 2.4 gives the indicated isomorphism on residue characters, with its alternating relation.

As long as the residual transcendence degree is at least two, apply Proposition 2.7 again there. The inverse-image identities (2.5) identify the resulting pairs with those of compositions on the original fields. Lemma 2.8 shows that each such inertia determines its composite valuation, so this matching is independent of any choice of a presentation as a flag. At each step Lemma 2.2 supplies finitely generated residue fields over algebraically closed residue constants. The local results apply even if one of these fields has characteristic ℓ . Iteration ends after $d - 1$ steps, with function fields of curves. $\square$

3 Recognizing points on residual curves

Corollary 2.9 identifies the residue character spaces at quasi-prime $(d - 1)$ -divisors. These residue fields have transcendence degree one, but their point valuations are not yet distinguished inside their character spaces. We now recognize those terminal residues whose constant fields are algebraic closures of finite fields. For these residues we also recognize every point inertia line. The construction uses Pop’s density theorem for an individual field, followed by reduction of its characters modulo ℓ .

Fix a function field F/κ over an algebraically closed field of characteristic different from ℓ , of transcendence degree $d \geq 2$, and let v be a quasi-prime $(d - 1)$ -divisor. Put

$$F_0 = Fv, \quad \kappa_0 = \kappa v.$$

By Lemma 2.4, restriction to units induces a surjective map $\rho_v : D_v \rightarrow A_{F_0}$ with kernel I_v . Define

$$S_F^{\text{in}} = \overline{\bigcup_{w \text{ quasi-prime divisor}} I_w} \subset A_F, \quad J_v = \rho_v(S_F^{\text{in}} \cap D_v) \subset A_{F_0}. \quad (3.1)$$

Here the bar denotes the closure of the union, rather than the subgroup generated by that union. In particular J_v is a set of characters, not in general a subspace. It is stable under multiplication by Λ , so it makes sense to consider the family of lines contained in J_v . Both sets in (3.1), and this family of lines, are preserved by the isomorphisms already constructed in Section 2.

3.1 The density input and reduction of coefficients

A valuation c of a function field B/b , with b algebraically closed, is a *constant reduction* if $\text{trdeg}(Bc/bc) = \text{trdeg}(B/b)$. The composition of a constant reduction with a prime divisor of Bc/bc is called a *c.r. quasi-prime divisor*. The trivial constant reduction is allowed, so ordinary prime divisors belong to this class. For a constant reduction, Lemma 2.2 gives $cB = cb$: the relative value group is torsion free of rational rank zero. After composition with a prime divisor, the relative value group is therefore $\mathbb{Z}$. The final discrete step is an innermost convex subgroup of the composed value group. Every proper coarsening kills this step and has all its remaining values supplied by constants. This proves the minimality required of a quasi-prime divisor and justifies the terminology.

For the density statement, temporarily use ℓ -adic characters:

$$\hat{A}_E = \text{Hom}(E^\times, \mathbb{Z}_\ell), \quad \hat{I}_a = \text{Hom}(E^\times/U_a, \mathbb{Z}_\ell), \quad \hat{D}_a = \text{Hom}(E^\times/U_a^1, \mathbb{Z}_\ell).$$

These groups have the topology of pointwise convergence. Restriction to units and passage to residue give a map $\hat{\rho}_v : \hat{D}_v \rightarrow \hat{A}_{F_0}$. Define $\hat{S}_F^{\text{in}}$ and $\hat{J}_v$ by the same formulas as in (3.1), with hats on the character groups and on ρ_v .

Theorem 3.1 (Pop's residual density input). *Let F/κ be a function field over an algebraically closed field of characteristic different from ℓ . Suppose that a valuation v of F , with $Fv/\kappa v$ a function field, has the following properties:*

- (i) *vF has no nonzero ℓ -divisible convex subgroup;*
- (ii) *there is a subfield $\kappa \subset \kappa_1 \subset F$ such that*

$$\kappa_1 v = \kappa v, \quad \text{trdeg}(F/\kappa_1) = \text{trdeg}(Fv/\kappa_1 v) = 1.$$

Then $\hat{J}_v$ contains the full minimized inertia group $\hat{I}_u$ of every c.r. quasi-prime divisor u of $Fv/\kappa v$.

This is the consequence of [8, Theorem 1.2 and the proof of Theorem 4.2, p. 351, item 2] with the notation introduced on p. 348. The auxiliary field κ_1 is not required to be algebraically closed. The minimized interpretation in residue characteristic ℓ is explained in [8, Remark 4.1 and the appendix, Section A].

We verify the hypotheses for our terminal valuation v . Lemma 2.8 shows that vF has no nonzero ℓ -divisible convex subgroup. It remains to construct the auxiliary field κ_1 .

Choose elements $x_1, \dots, x_{d-1} \in F^\times$ whose values are rationally independent modulo $v\kappa$, and set $\kappa_1 = \kappa(x_1, \dots, x_{d-1})$. Distinct monomials in the x_i have different values modulo $v\kappa$. Thus every nonzero polynomial in these elements has a unique term of least value, proving algebraic

independence over κ . If a quotient of two such polynomials has value zero, their least terms have the same monomial. Its residue is the residue of the quotient of their coefficients, and consequently belongs to κv . This proves $\kappa_1 v = \kappa v$; the two transcendence degrees required by Theorem 3.1 are now one.

Lemma 3.2. *For a quasi-prime $(d-1)$ -divisor v of F/κ , the set J_v contains the inertia line $I_u \subset A_{F_0}$ of every c.r. quasi-prime divisor u of F_0/κ_0 .*

Proof. Reduction of character values gives continuous maps

$$r_E : \hat{A}_E \longrightarrow A_E.$$

They carry $\hat{I}_a$ into I_a , carry $\hat{D}_a$ into D_a , and commute with restriction to units. In particular,

$$r_F(\hat{S}_F^{\text{in}}) \subset S_F^{\text{in}}, \quad \rho_v r_F = r_{F_0} \hat{\rho}_v \quad \text{on } \hat{D}_v.$$

The first inclusion follows from continuity and the definition as a closure of a union.

Fix a c.r. quasi-prime divisor u of F_0/κ_0 . Characters with either coefficient ring annihilate the divisible group $u\kappa_0$. Since $uF_0/u\kappa_0 \cong \mathbb{Z}$, reduction $\hat{I}_u \rightarrow I_u$ is surjective. Given $\chi \in I_u$, choose a lift $\hat{\chi} \in \hat{I}_u$. Theorem 3.1 supplies $\hat{\eta} \in \hat{S}_F^{\text{in}} \cap \hat{D}_v$ with $\hat{\rho}_v(\hat{\eta}) = \hat{\chi}$. Then $r_F(\hat{\eta})$ belongs to $S_F^{\text{in}} \cap D_v$ and has residual image χ . Hence $\chi \in J_v$. $\square$

The argument needs only the displayed inclusion between the two closed unions. It does not assert that reduction commutes with their intersection with decomposition groups, and it makes no use of a lift of Θ .

3.2 Valuative characters and the relation among point orders

The density input gives a lower bound for J_v . To control its other elements, we use the fact that taking this closed union introduces no nonvaluative characters.

Lemma 3.3. *Every element of S_F^{in} is valuative. Every element of J_v is valuative as a character of F_0 .*

Proof. Let $\mathcal{V}(F)$ be the space of valuation rings of F , with its compact patch topology. In $A_F \times \mathcal{V}(F)$, the condition that a character χ annihilates the unit group of a valuation ring O is closed. Indeed, for each $x \in F^\times$, failure of this condition is witnessed by the open conditions

$$x, x^{-1} \in O, \quad \chi(x) \neq 0.$$

The projection of the closed incidence set to A_F is compact, hence closed. This projection is precisely the set of valuative characters, which contains every I_w and therefore contains S_F^{in} .

Now let $\chi \in I_u \cap D_v$ for some valuation u . If $u \leq v$, then $\chi \in I_v$ and its residual image is zero. If $v \leq u$, then restriction to U_v shows that $\rho_v(\chi)$ annihilates the units of the induced valuation u/v on F_0 . In the remaining case let a be the finest common coarsening of u and v . Approximation for the independent induced valuations on Fa gives

$$U_a = U_u U_v^1.$$

For example, a prescribed nonzero residue can be written as an induced u -unit times an induced principal v -unit by choosing an element close to 1 at the first valuation and close to that residue at the second. Lifting gives the displayed identity; the remaining factor in U_a^1 is already a unit for both refinements. Thus χ annihilates U_a , and $\chi \in I_a \subset I_v$. Its residual image is again zero. $\square$

We next record how the points of a complete curve appear in its character space, in the finite-coefficient form of Topaz's calculation in [8, appendix, Lemma 5, pp. 354–355]. This description will also be used for curve subfields later.

Lemma 3.4 (Point orders). *Let B/b be a function field of transcendence degree one over an algebraically closed field, and let C/b be its smooth projective curve. For $x \in C(b)$, let $\delta_x = \text{ord}_x \bmod \ell \in A_B$. The lines $\Lambda\delta_x$ are distinct and nonzero. The family $(\delta_x)_{x \in C(b)}$ tends to zero outside finite subsets, and the continuous summation map*

$$\prod_{x \in C(b)} \Lambda \longrightarrow A_B, \quad (a_x)_x \longmapsto \sum_x a_x \delta_x \quad (3.2)$$

has kernel consisting exactly of the constant families. If $\text{char}(b) \neq \ell$, the common kernel in V_B of the δ_x has dimension $2g(C)$.

Proof. Approximation at distinct point valuations proves that each δ_x is nonzero and that their lines are distinct. A function on C has only finitely many zeros and poles. Hence every fixed element of $B^\times$ is annihilated by all but finitely many δ_x , which proves the asserted convergence and defines (3.2), with continuous evaluation at each function.

A family $(a_x)_x$ defines a homomorphism from the divisor group of C to Λ , sending the point divisor x to a_x . It lies in the kernel of (3.2) precisely when this homomorphism vanishes on principal divisors, or equivalently factors through $\text{Pic}(C)$. The group $\text{Pic}^0(C)(b)$ is ℓ -divisible: multiplication by ℓ on the Jacobian is a surjective isogeny, and b is algebraically closed. This remains true in characteristic ℓ . Every homomorphism $\text{Pic}(C) \rightarrow \Lambda$ therefore factors through the degree map $\text{Pic}(C) \rightarrow \mathbb{Z}$. Since all point divisors have degree one, its coefficient family is constant. Conversely, constant families annihilate principal divisors.

Finally, if all point orders of $f \in B^\times$ are divisible by ℓ , write $\text{div}(f) = \ell D$ and associate to $[f]$ the class of D in $\text{Pic}(C)[\ell]$. This gives an isomorphism from the common kernel to $\text{Pic}(C)[\ell]$: surjectivity follows from the definition of torsion in $\text{Pic}(C)$; injectivity follows because $b^\times$ is ℓ -divisible. When $\text{char}(b) \neq \ell$, the ℓ -torsion of the Jacobian is isomorphic to $(\mathbb{Z}/\ell)^{2g(C)}$. $\square$

3.3 An intrinsic test for finite-field constants

We now adapt the curve-like relation criterion of [8, Theorem 4.2 and appendix, Main Theorem I, p. 353] to the family of lines in J_v . Let $\mathcal{L}_v$ be the family of one-dimensional subspaces of A_{F_0} contained in J_v . Choose a nonzero generator $e_H \in H$ for each $H \in \mathcal{L}_v$. Consider the following conditions:

- (i) the family $(e_H)_{H \in \mathcal{L}_v}$ tends to zero outside finite subsets;
- (ii) the resulting summation map

$$\prod_{H \in \mathcal{L}_v} \Lambda \longrightarrow A_{F_0}, \quad (a_H)_H \longmapsto \sum_H a_H e_H$$

has a one-dimensional kernel generated by a family whose every coordinate is nonzero.

These conditions are independent of the choices of generators. Changing generators rescales the coordinates of the product, and, because Λ is finite, preserves the convergence in the first condition. They are also preserved by topological linear isomorphisms of the residual character spaces.

Proposition 3.5. *For a quasi-prime $(d-1)$ -divisor v of F/κ , the family $\mathcal{L}_v$ satisfies the preceding two conditions if and only if κv is algebraic over a finite field. When this holds, $\mathcal{L}_v$ consists exactly of the point inertia lines of the smooth projective curve of $Fv/\kappa v$. Consequently the matching in Corollary 2.9 recognizes these terminal valuations and matches their full point families.*

Proof. Suppose first that κ_0 is algebraic over a finite field. Every valuation on κ_0 is trivial. A nontrivial valuation of the one-variable field F_0 trivial on κ_0 is a point valuation on its smooth projective curve: properness gives a closed center, and the discrete valuation ring at that smooth point is dominated by the valuation ring. Writing each function as a power of a uniformizer times a local unit identifies the two valuations. Thus Lemma 3.3 shows that every line in $\mathcal{L}_v$ is a point line. The converse inclusion follows from Lemma 3.2, using the trivial constant reduction. Lemma 3.4 now proves the two conditions.

Suppose instead that κ_0 is not algebraic over a finite field. There is a nontrivial valuation b on κ_0 : in characteristic zero extend a nontrivial valuation of the prime field; in positive characteristic use a transcendental element, extend its variable valuation to a rational transcendence basis, and prolong to κ_0 . Choose a transcendental $t \in F_0$ with $F_0/\kappa_0(t)$ finite. The Gauss extension of b to $\kappa_0(t)$, followed by a prolongation to F_0 , gives a constant reduction c . Indeed the rational residue field has transcendence degree one, and finite prolongation gives a finite residue extension. The value group of c equals its constant-value subgroup: the Gauss extension has this property, and a finite prolongation has finite value-group index, whereas the constant-value group is divisible.

Choose a point valuation q on F_0c/κ_0c , and form $u = q \circ c$. This is a c.r. quasi-prime divisor, whose value group contains the final discrete step as an innermost convex subgroup. In particular $uF_0/u\kappa_0 \cong \mathbb{Z}$, so I_u is a nonzero line, and it belongs to $\mathcal{L}_v$. It is different from every true point line. To see this, let p be a point valuation of F_0/κ_0 . The valuation u is nontrivial on constants, so cannot coarsen p . Conversely, a proper coarsening of u kills its innermost discrete subgroup and retains only values supplied by constants; it cannot equal the nontrivial valuation p that is trivial on constants. Thus p and u are incomparable. Since p has rank one, their finest common coarsening is trivial. Approximation gives $I_p \cap I_u = 0$, proving the assertion even with Λ -coefficients.

All point lines already belong to $\mathcal{L}_v$, and their generators have the nonzero relation of Lemma 3.4, after rescaling to the chosen generators. If the first condition holds for the full family, extend this relation by zero on the additional lines. It is a nonzero kernel vector with a zero coordinate, contradicting the second condition. If the first condition fails, the family also fails the stated test. This proves the equivalence. $\square$

4 A uniform bound for the support of a pencil

Fix a normal integral projective model $X \subset \mathbb{P}_\ell^n$ of L/l . For $h \in V_L$, define

$$\text{supp}_X(h) = \{D \subset X : D \text{ is a prime divisor and } \text{ord}_D(h) \neq 0 \text{ in } \Lambda\}.$$

Its degree is

$$s_X(h) = \sum_{D \in \text{supp}_X(h)} \deg(D).$$

The orders modulo ℓ , and hence this finite set and its degree, depend only on the class h . We will prove that, for each $t \in K \setminus k$, the integers $s_X(\Theta([t-a]))$ are bounded independently of $a \in k$. The method is to realize any finite selection of these classes on a residual curve of

bounded genus. The point correspondence from Section 3 then turns the desired bound into a Riemann–Hurwitz estimate. The bounded-genus and Hurwitz method appears in [4, Section 6, proof of Proposition 6.1]; we adapt the bounded-support construction of [6, Sections 4–6] to finite coefficients.

4.1 Finite Kummer tests

We first record why only finitely many classes can disappear in a finitely generated extension.

Lemma 4.1. *Let E/F be a finitely generated field extension, where $\text{char } F \neq \ell$ and $\mu_\ell \subset F$. The kernel of $V_F \rightarrow V_E$ is finite dimensional over Λ .*

Proof. Let F' be the relative algebraic closure of F in E . Choose a transcendence basis $\mathbf{x}$ for E/F such that $[E : F(\mathbf{x})] = N < \infty$. For every finite extension F_1/F contained in F' , the tuple $\mathbf{x}$ remains algebraically independent over F_1 , and

$$[F_1 : F] = [F_1(\mathbf{x}) : F(\mathbf{x})] \leq N.$$

The finite subextensions of F'/F form a directed system with bounded degrees. One of them has maximal degree and contains all the others; thus F'/F is finite.

If r independent classes in V_F vanish in V_E , choose representatives $f_1, \dots, f_r \in F^\times$ and their ℓ -th roots in E . Those roots belong to F' , and Kummer theory gives

$$[F(f_1^{1/\ell}, \dots, f_r^{1/\ell}) : F] = \ell^r.$$

Hence $\ell^r \leq [F' : F]$, which bounds the kernel dimension. $\square$

The following geometric observation explains why a single smooth curve can preserve any specified finite Kummer extension. Its irreducibility assertion is useful because the cover can have arbitrarily large degree.

Lemma 4.2. *Let U be a smooth integral locally closed subvariety of projective space over an algebraically closed field, of dimension at least two. Let $Y \rightarrow U$ be a finite étale morphism with Y integral. For general hyperplanes H , both $U \cap H$ and $Y \times_U (U \cap H)$ are nonempty, smooth, and integral.*

Proof. Smoothness of $U \cap H$ is Bertini’s smoothness theorem for an embedded smooth variety, valid in every characteristic [11, Lemma 33.47.3, Tag 0FD6]. It implies smoothness upstairs because the cover is étale. We give the irreducibility argument, applied to either $Y \rightarrow \mathbb{P}^n$ or $U \rightarrow \mathbb{P}^n$.

Write $f : Y \rightarrow \mathbb{P}^n$ for the resulting quasi-finite morphism and $a = \dim Y \geq 2$. Let $\mathcal{H} = (\mathbb{P}^n)^\vee$ be the space of hyperplanes. In the incidence of triples (y_1, y_2, H) with $f(y_1), f(y_2) \in H$, the locus $f(y_1) \neq f(y_2)$ is a projective space bundle with fiber $\mathbb{P}^{n-2}$ over a nonempty open of $Y \times Y$. It is therefore irreducible of dimension $2a + n - 2$. The locus of pairs with equal images has dimension at most a , by quasi-finiteness; its hyperplane incidence has dimension at most $a + n - 1 < 2a + n - 2$.

For a general hyperplane, $f^{-1}(H)$ is nonempty and has pure dimension $a - 1$. Nonemptiness follows because the image of f contains an open of its a -dimensional closure; purity follows from the principal ideal theorem on the integral variety Y . Thus every component of the square of the generic hyperplane section has dimension $2a - 2$ over the function field of $\mathcal{H}$. No such component

can be contained in the equal-image incidence, whose total dimension is too small. The square of the generic section is consequently irreducible.

This implies geometric irreducibility of the generic section. Indeed, if it had more than one geometric component, ordered pairs of points lying on the same component and on different components would give two distinct Galois-invariant unions of components of its square. Its already established generic smoothness excludes nonreducedness. Geometric integrality spreads to a nonempty open of $\mathcal{H}$, which proves the assertion for general hyperplanes. $\square$

Proposition 4.3 (Finite curve test). *There is an integer $G_X \geq 0$, depending only on the embedded model X , with the following property. Given $g_1, \dots, g_m \in L^\times$ whose classes in V_L are independent, there is a quasi-prime $(d-1)$ -divisor w of L/l such that:*

- (i) lw is algebraically closed and algebraic over a finite field of characteristic different from ℓ , and Lw is the function field of a smooth projective curve C_w/lw of genus at most G_X ;
- (ii) the g_j are w -units and their residue classes are independent in V_{Lw} ;
- (iii) for every j ,

$$\#\{x \in C_w(lw) : \text{ord}_x(\overline{g_j}) \not\equiv 0 \pmod{\ell}\} = s_X([g_j]).$$

Proof. We first construct a curve over l , then specialize its finite defining data, and finally realize that specialization by a valuation of L .

A curve preserving orders and independence. Let $D_1, \dots, D_q$ be the prime divisors occurring in the divisors of the g_j . There is a closed subset $B \subset X$ of codimension at least two such that $X \setminus B$ is smooth, the $D_i \setminus B$ are smooth pairwise disjoint Cartier divisors, and near each $D_i \setminus B$ every g_j is a power of a local equation of D_i times a unit. Outside their union the functions are units. To obtain B , remove the singular locus of the normal variety X , the singular and non-Cartier loci of the D_i , their pairwise intersections, and the exceptional loci of these local expressions. Each has codimension at least two.

On the smooth open $U = X \setminus (B \cup D_1 \cup \dots \cup D_q)$, the simultaneous root cover defined by the equations $z_j^\ell = g_j$ is finite étale. Kummer theory and independence say that it is integral of degree ℓ^m . Successively choose $d-1$ general hyperplanes. Apply Lemma 4.2 to the cover on each successive open section, and Bertini smoothness to the base and its specified divisors. We obtain a smooth integral projective curve $C \subset X$ that avoids B , meets every D_i transversely in exactly $\deg D_i$ distinct points, and has an integral root cover over $C \cap U$. The latter cover still has degree ℓ^m , so the restricted classes of the g_j are independent. At a point of $C \cap D_i$ their orders are exactly their orders along D_i . These are all their zeros and poles on C , proving the required equality of support counts over l .

The genus of this curve is independent of the chosen functions. At each stage we may also require the hyperplane to avoid the associated points of the preceding scheme section. Multiplication by its equation is then injective, so the Hilbert polynomial of a section is the first difference of the preceding Hilbert polynomial. The final scheme section avoids B , and on $X \setminus B$ the successive sections are smooth by Bertini. Hence the final scheme section is reduced and equals C , even if intermediate sections had embedded components supported in B . Its Hilbert polynomial, and thus its genus, depend only on the Hilbert polynomial of the embedded X . Denote this genus by G_X . This argument does not require X to be Cohen–Macaulay.

Specialization of the finite data. Choose a finitely generated subring $R \subset l$ in which ℓ is invertible and over which the preceding data are defined. Enlarge R by finitely many elements and localize it as follows. The smooth open of X containing C , the flag sections in that open, and

their inclusions descend with their dimensions, smoothness, and geometric integrality preserved in every geometric fiber. The final curve descends as a smooth projective curve of genus G_X . Include the coordinates of its finitely many zeros and poles and the local parameter–unit expressions for the functions. After shrinking $\text{Spec } R$, they give disjoint point sections with the same orders and no other zeros or poles. The root cover over the complement of those sections remains finite étale of degree ℓ^m , with geometrically integral fibers. Thus independence and the support counts persist in every geometric fiber under consideration.

Here the spreading statements have their usual finite-presentation meaning: finitely many schemes, morphisms, functions, and identities descend after adjoining finitely many coefficients [11, Tags 01ZM, 0C0C, 081F]; smoothness and flatness hold after shrinking; and geometric integrality at the generic point persists on an open [11, Tags 0578, 0559]. The Hilbert polynomial in the projective flat family of curves is constant; its constant term is the fiberwise Euler characteristic, locally constant in this proper flat family [11, Lemma 36.32.2, Tag 0B9T]. For the orders, the identities $g_j = u\pi^e$ on finitely many neighborhoods, with u invertible and π a parameter for the relevant section, preserve the integer e . On the complement of these neighborhoods the functions and their inverses are regular. Properness of the curve ensures that these finitely many open conditions still cover every fiber after shrinking.

We also include a presentation of the function field, so that the specialized variety will be the residue field of a valuation. Choose a separating transcendence basis $x_1, \dots, x_d$ for L/l and a primitive element y for the finite separable extension $L/l(x_1, \dots, x_d)$. Write its monic minimal polynomial as $Q(Y) \in l(\mathbf{x})[Y]$. After inverting a polynomial in $\mathbf{x}$, this gives a finite integral model over an open of affine d -space. Descend that model, a common dense open with the model already chosen, and the rational expressions for all the g_j . By geometric integrality and further localization of R , the reduced polynomial in every geometric fiber is irreducible of the same degree as Q and gives that fiber's function field. All denominators in the coefficients and in the expressions being used remain nonzero, and the g_j agree with the specified nonzero functions there.

Realization by a valuation. Choose a closed point $s \in \text{Spec } R$. Its residue field is finite, of characteristic different from ℓ . A valuation of $\text{Frac } R$ dominating R_s extends to l ; equivalently, one may directly choose a valuation ring of l dominating R_s [11, Lemma 10.50.2, Tag 00IA]. Write its residue field as λ_0 . Since l is algebraically closed, so is λ_0 . We may arrange that the ultimate residue field is algebraic over the finite field $\kappa(s)$. Indeed let b be the algebraic closure of $\kappa(s)$ inside λ_0 , and choose a transcendence basis $\mathcal{T}$ for λ_0/b . On $b(\mathcal{T})$ give the basis elements rationally independent values in an ordered free abelian group with basis $\mathcal{T}$ and give $b^\times$ value zero. Every polynomial has a unique term of least value, so this defines a valuation with residue b . Extend it to the algebraic extension $\lambda_0/b(\mathcal{T})$. Its residue extension is algebraic and thus still equals b . Composing valuations gives a valuation u of l with residue $\lambda = b$. Its center on R remains s , since the second valuation is trivial on the finite field $\kappa(s)$. An ordered free abelian group exists for any cardinality of $\mathcal{T}$, so this construction imposes no cardinality condition on l .

Give $l(\mathbf{x})$ the Gauss extension of u : the values of the x_i are zero and their residues are algebraically independent over λ . Prolong it to L , obtaining w_X . The coefficients of Q are integral at the Gauss valuation and its reduction is the specified irreducible polynomial over $\lambda(\overline{\mathbf{x}})$. Since Q is monic, y is integral; its residue therefore has degree $\deg Q$. The fundamental inequality for a finite extension of valued fields now forces

$$w_X L = ul, \quad Lw_X = \lambda(\overline{\mathbf{x}}, \overline{y}).$$

Thus w_X is a constant reduction with precisely the specialized function field as residue, and each

g_j has the prescribed residue.

On Lw_X , take the successive true divisor valuations of the specialized smooth flag, ending at its curve. Compose them with w_X . Write w_1 for the first composition. It is quasi-divisorial: its value group Γ has a convex kernel $\mathbb{Z}$ over $w_X L$, while the constant values map isomorphically onto $w_X L = ul$. Consequently $\Gamma/w_1 l \cong \mathbb{Z}$. Every nonzero convex subgroup of Γ contains this innermost $\mathbb{Z}$, so any proper coarsening has value group generated by constant values and loses the relative rank-one contribution. This proves the required minimality. The remaining steps are true divisor steps. Their composition w is therefore a quasi-prime $(d-1)$ -divisor with residue the specialized curve field. All g_j are units at the successive generic points. The genus, independence, and support counts already preserved in the specialization give the three conclusions. $\square$

4.2 Five fibers control the degree

We now transfer the finite tests across Θ . The curve used for a test is allowed to depend on the tested classes; its genus bound depends only on X .

Proposition 4.4. *For every $t \in K \setminus k$,*

$$\sup_{a \in k} s_X(\Theta([t-a])) < \infty.$$

Proof. Put $h_a = \Theta([t-a])$. The classes $[t-a]$, $a \in k$, are independent in $V_{k(t)}$, as is seen from their orders at the finite points of the projective line. Lemma 4.1 shows that the kernel of $V_{k(t)} \rightarrow V_K$ is finite dimensional. A basis for its intersection with the span of these classes involves only finitely many indices. After removing those indices, we have an infinite set $A \subset k$ such that the classes $[t-a]$, $a \in A$, are independent in V_K .

Choose five distinct anchors $a_1, \dots, a_5 \in A$ and put $N_i = s_X(h_{a_i})$. If $\text{char } k = 0$, choose the anchors to include $b_0, b_0 + 1, b_0 + \ell$; this is possible because $k \setminus A$ is finite. Fix one further $a \in A$ distinct from the anchors and put $a_6 = a$. Choose representatives $g_1, \dots, g_6 \in L^\times$ of $h_{a_1}, \dots, h_{a_6}$, and apply Proposition 4.3 to these representatives. Let w be the resulting valuation and C_w its residual curve. Their residue classes are independent, so their evaluations on D_w are independent by Lemma 2.4. Let v be the quasi-prime $(d-1)$ -divisor of K corresponding to w under Corollary 2.9. The evaluations of the six functions $u_j = t - a_j$, $1 \leq j \leq 6$, on D_v are independent.

We first normalize these functions so that their residues are a pencil on Kv . We claim that $v(k(t)) = vk$. Otherwise the nonzero relative value group of $k(t)/k$ has rational rank one, and the valuation transcendence-degree inequality forces $k(t)v = kv$. Equality in that inequality makes the relative value group cyclic. Any element with value in vk can then be multiplied by a constant to become a unit, and its residue can be removed by a further constant. Its class consequently vanishes on D_v . Thus evaluation of $k(t)^\times$ on D_v factors through the cyclic relative value group and has dimension at most one, a contradiction.

The values of all six u_j are equal. If $v(u_i) > v(u_j)$, then $u_j/(a_i - a_j)$ is a principal unit, contradicting its nonzero evaluation on D_v . Write γ for the common value. For distinct indices, $v(a_i - a_j) \geq \gamma$; a strict inequality would make u_i/u_j a principal unit and give equal evaluations. Hence every difference $a_i - a_j$ has value exactly γ . In positive characteristic the residue characteristic equals $\text{char } k \neq \ell$. In characteristic zero our three specified anchors give $v(1) = v(\ell) = \gamma = 0$, so again $\text{char}(kv) \neq \ell$.

Choose $b \in k^\times$ with $v(b) = \gamma$ and set

$$\tau_0 = \overline{(t - a_1)/b}, \quad \beta_j = \overline{(a_j - a_1)/b}.$$

The $\beta_j \in kv$ are distinct and $\overline{u_j/b} = \tau_0 - \beta_j$. These residue classes remain independent: the functions u_j/b are units, their evaluations factor through D_v/I_v , and multiplication by b changes no class because constants are ℓ -divisible. In particular $\tau_0 \notin kv$. More precisely, the dual of the induced isomorphism $D_w/I_w \rightarrow D_v/I_v$ carries $[\tau_0 - \beta_j] \in V_{Kv}$ to $[\overline{g_j}] \in V_{Lw}$. Indeed their evaluations on corresponding residual characters are the evaluations of $[u_j]$ and $\Theta([u_j]) = [g_j]$ on the original decomposition spaces. This identifies the individual tested classes, as well as preserving their independence.

Proposition 3.5 says that kv , like lw , is algebraic over a finite field, and that the complete point families on the two residual curves correspond. Let C_v be the smooth projective curve with function field Kv . Both residue characteristics are different from ℓ ; by Lemma 3.4, the dimensions of the common point-order kernels are twice the genera. The residual isomorphism preserves these kernels, so $g(C_v) = g(C_w) \leq G_X$. It also preserves the number of nonzero point orders of each tested class. For the five anchor classes these numbers are $N_1, \dots, N_5$.

Write $p = \text{char}(kv) > 0$ and $\tau_0 = \tau^{p^r}$ with r maximal. Such an r exists because a nonconstant function has a nonzero point order, and p^r must divide that fixed nonzero integer. The function τ defines a separable morphism $C_v \rightarrow \mathbb{P}_{kv}^1$; write n' for its degree. Multiplication of orders by p^r does not change their vanishing modulo ℓ .

Consider the fiber of this morphism over β_i^{1/p^r} . At most N_i points in the fiber have ramification index not divisible by ℓ . The other points have index at least ℓ , so there are at most n'/ℓ of them. If r_i is the total number of points in the fiber, then $r_i \leq N_i + n'/\ell$. Since the sum of the ramification indices in a fiber is n' , and the different exponent at a point is at least its index minus one, the fiber contributes at least $n' - r_i \geq (1 - 1/\ell)n' - N_i$ to the different. The five fibers are disjoint. Riemann–Hurwitz, with its different term in arbitrary characteristic [11, Lemma 53.12.4, Tag 0C1F], therefore yields

$$2g(C_v) - 2 + 2n' \geq 5(1 - 1/\ell)n' - \sum_{i=1}^5 N_i,$$

or

$$(5(1 - 1/\ell) - 2)n' \leq 2G_X - 2 + \sum_{i=1}^5 N_i. \quad (4.1)$$

The coefficient on the left is positive for every prime ℓ , including $\ell = 2$. Hence n' has a bound depending only on the five anchors and X .

For the additional parameter, the zeros and poles of $\tau_0 - \beta_6$ lie in two fibers of this same morphism. It has at most $2n'$ nonzero point orders modulo ℓ . The point correspondence and the finite curve test identify this number with $s_X(h_a)$. This proves a uniform bound for all non-anchor members of A . Including the anchors and the finitely many parameters outside A completes the proof. $\square$

5 Recovering curve subfields

The support bound of Proposition 4.4 allows us to pass from individual multiplicative classes to subfields. We prove that Θ carries the mod- ℓ multiplicative group of every relatively algebraically closed one-variable subfield of K onto that of such a subfield of L . The geometric step is an incidence construction: infinitely many divisors of bounded degree form a family, and a curve in its parameter space produces the one-variable field. This follows the bounded-support and

incidence method of [6, Sections 4–6]. We give the construction and the descent argument in the finite-coefficient setting.

A *curve subfield* of a function field F/κ is a subfield $E \subset F$ containing κ , relatively algebraically closed in F , and of transcendence degree one over κ . It is finitely generated: for any $t \in E \setminus \kappa$, it is the relative algebraic closure of $\kappa(t)$ in F , which is finite over $\kappa(t)$. The natural map $V_E \rightarrow V_F$ is injective, since an ℓ -th root in F of an element of E is algebraic over E . We henceforth regard V_E as a subspace of V_F .

5.1 Two descent facts

We first record the field-theoretic facts needed to descend the field produced by incidence. Recall that a finitely generated extension F/B is *regular* if it is separable and B is relatively algebraically closed in F ; equivalently, F and an algebraic closure $\overline{B}$ are linearly disjoint over B .

Lemma 5.1 (Descent from algebraically closed constants). *Let F/B be a finitely generated regular extension, where $\text{char}(B) \neq \ell$ and $\mu_\ell \subset B$. If $g \in F^\times$ has an ℓ -th root in $F\overline{B}$, then its class in V_F lies in the image of V_B .*

Proof. Choose $r \in F\overline{B}$ with $r^\ell = g$. The element r already belongs to FB' for a finite Galois extension B'/B . Indeed it is separable over F , so purely inseparable constant extensions are unnecessary, and we may take a finite Galois closure of the remaining constant extension. By regularity, $\text{Gal}(FB'/F) = \text{Gal}(B'/B)$. For γ in this group,

$$c_\gamma = \frac{\gamma(r)}{r} \in \mu_\ell.$$

These multipliers form a multiplicative cocycle with values in $B'^\times$. Hilbert's Theorem 90 gives $b' \in B'^\times$ with $\gamma(b')/b' = c_\gamma$ for every γ . Thus $r/b' \in F$ and $b'^\ell \in B$. The equality $g = (r/b')^\ell b'^\ell$ proves the assertion. $\square$

Lemma 5.2 (Intersections of curve subfields). *Let F/κ be a finitely generated extension of an algebraically closed field of characteristic different from ℓ . If $P, Q \subset F$ are distinct curve subfields, then*

$$\dim_\Lambda(V_P \cap V_Q) < \infty.$$

Each V_P is infinite dimensional. In particular, an inclusion $V_P \subset V_Q$ forces $P = Q$.

Proof. The compositum PQ has transcendence degree two over κ . Otherwise every element of Q would be algebraic over P , so relative algebraic closedness of P in F would give $Q \subset P$; reversing their roles would give equality. Let C_P, C_Q be their smooth projective curves. The natural map to $C_P \times C_Q$ is dominant, since its image has dimension two; consequently PQ is the function field of this product.

Inside V_{PQ} , any class coming from both fields has zero order along every divisor $\{x\} \times C_Q$. Its representative from P therefore lies in the common kernel of all point orders of C_P . This kernel has dimension $2g(C_P)$ by Lemma 3.4, so the intersection inside V_{PQ} is finite dimensional. The maps from V_P and V_Q to V_{PQ} are injective: each factor field is relatively algebraically closed in the function field of the product.

The kernel of $V_{PQ} \rightarrow V_F$ is finite dimensional by Lemma 4.1. To see that passing to F preserves the finite-intersection conclusion, consider pairs $(a, b) \in V_P \oplus V_Q$ whose images in V_F agree. The difference $a - b$ lies in this finite kernel; the kernel of the difference map on such pairs

is the intersection already computed in V_{PQ} . The space of these pairs, and hence $V_P \cap V_Q$ in V_F , is finite dimensional.

Finally, choose $t \in P \setminus \kappa$. The classes $[t - a]$, $a \in \kappa$, are linearly independent in $V_{\kappa(t)}$, as their orders at the distinct finite points show. Lemma 4.1 gives only a finite-dimensional kernel on passing to V_P , so V_P is infinite dimensional. $\square$

5.2 A family of divisors and a field of constants

The next lemma isolates the geometric construction. Its hypothesis says that every class under consideration becomes an ℓ -th power on almost every divisor in one fixed infinite family. The conclusion realizes all these classes as classes from a single curve field after a finite extension of the ambient field.

Lemma 5.3 (Incidence descent). *Let $X \subset \mathbb{P}_l^n$ be a normal integral projective variety of dimension $d \geq 2$ over an algebraically closed field l of characteristic different from ℓ , and put $L = l(X)$. Let $\mathcal{S}$ be an infinite set of prime divisors on X whose degrees are bounded. Suppose that $H \subset V_L$ is a subspace with the following property: for every $g \in L^\times$ with $[g] \in H$,*

$$\text{ord}_D(g) = 0 \quad \text{and} \quad g|_D \in l(D)^{\times \ell} \quad \text{for all but finitely many } D \in \mathcal{S}. \quad (5.1)$$

Then there are a finite extension M/L and a one-variable field P_0/l contained in M such that M/P_0 is regular,

$$M = LP_0, \quad (5.2)$$

and the image of H in V_M is contained in the image of V_{P_0} . In particular, every g as above has an ℓ -th root in $M\overline{P_0}$.

Proof. *A bounded parameter space.* We recall why a degree bound gives a parameter space of finite type. Integral subvarieties of fixed dimension and bounded degree in a fixed projective space have only finitely many Hilbert polynomials; see also [10, Proposition 5.3]. One way to establish this boundedness is to cut them out set-theoretically by forms of bounded degree. If an integral subvariety $D \subset \mathbb{P}^n$ has dimension r and degree at most b , and $x \notin D$, choose a linear projection to $\mathbb{P}^{r+1}$ whose center avoids the join of x and D . When D is already a hypersurface no projection is necessary. The projection is defined on D and is finite there: a positive-dimensional fiber would contradict ampleness of the pulled-back hyperplane bundle. Its image is a hypersurface of degree at most b , and its equation pulls back to a form vanishing on D but not at x . Multiplication by a form nonzero at x makes the degree exactly b if necessary. It follows that the degree- b forms vanishing on D cut it out as a reduced set.

A fixed number of such forms, namely the dimension of the space of degree- b forms, therefore places all these reduced subvarieties among the geometric reductions of fibers of one projective family of finite type. Such geometric reductions have finitely many Hilbert polynomials. For completeness, over an integral base descend the reduction of the geometric generic fiber to a finite extension of the base function field, and normalize the base in that extension. On a dense open of this finite base change, spread the reduction as a closed subscheme, with nilpotent defining ideal, flat and with geometrically reduced fibers. These properties follow after shrinking from generic flatness and spreading geometric reducedness in the resulting projective flat family [11, Tag 0578]. Thus it gives exactly the geometric reductions there, with constant Hilbert polynomial. The complement upstairs has image in a proper closed subset of the original base because the normalization is finite. Noetherian induction on that closed subset proves the assertion. Applying this to the divisors in $\mathcal{S}$ gives the required finite union of Hilbert schemes on X .

The incidence fields. In that union take a positive-dimensional irreducible component of the closure of the points corresponding to $\mathcal{S}$. After replacing it by an integral locally closed open subset T , the tested points are dense in T , and its universal family

$$Z \subset X \times T \longrightarrow T$$

is flat with geometrically integral fibers of dimension $d - 1$. The latter condition is available by constructibility of geometric integrality [11, Tags 0579 and 055B]: the tested fibers are integral over the algebraically closed field l , and their parameter points are dense. Choose an integral locally closed curve $T' \subset T$, once and for all, and put $Z' = Z \times_T T'$. Both Z and Z' are integral, by flatness and geometric integrality of their generic fibers. Their maps to X are dominant. Indeed a proper closed subset of X can contain only finitely many distinct prime divisors, whereas the fibers over the infinitely many distinct points of T , or of T' , are distinct divisors.

The two projections now give

$$\begin{array}{ccccc} & Z' & & M = l(Z') & \\ & \swarrow \quad \searrow & & \nwarrow \quad \swarrow & \\ X & & T' & L = l(X) & P_0 = l(T'). \end{array}$$

Since $\dim Z' = d$, the extension M/L is finite. Geometric integrality of the generic fiber of $Z' \rightarrow T'$ says precisely that M/P_0 is regular. The inclusion $Z' \subset X \times T'$ also shows that its function field is generated by the coordinates from the two factors, proving (5.2).

Descent along the fixed curve. We next show that the chosen fields work for every class in H . Fix $g \in L^\times$ with $[g] \in H$. Let Y be the intersection of the relative smooth locus of $Z \rightarrow T$ with the inverse image of the open subset of X on which g is a unit. Every nonempty geometric fiber of $Y \rightarrow T$ is integral. Its fibers over the generic points of T and T' are nonempty: the corresponding incidence families dominate X , and the smooth locus is dense in their geometrically integral fibers. On Y consider the finite étale cover defined by adjoining an ℓ -th root of g .

This cover splits on a dense set of the tested fibers. Indeed (5.1) gives a rational ℓ -th root on each such divisor, and on its smooth unit locus the root and its inverse are regular, by normality. The cover therefore splits on the geometric generic fiber over T as well. Otherwise, since ℓ is prime and the constants contain μ_ℓ , its generic Kummer polynomial would be irreducible over that geometric function field. The geometric generic cover would then be integral. Geometric integrality spreads to a nonempty open of T [11, Tags 0578 and 0559], contradicting the dense set of split fibers.

It remains to pass this splitting to our fixed curve T' . Shrinking T separately for each g would not justify this passage: the resulting open could miss T' . Instead, descend the geometric generic splitting to a finite extension of $l(T)$ and let $\tilde{T} \rightarrow T$ be the finite surjective normalization in that extension. The pullback $\tilde{Y} = Y \times_T \tilde{T}$ is normal, since it is smooth over the normal scheme $\tilde{T}$ [11, Tag 034F]. It is also integral. Its generic fiber is geometrically integral, and every irreducible component dominates $\tilde{T}$: on a normal scheme the components are open, and a smooth map has open image. Thus there is only one component.

The pulled-back finite étale cover of $\tilde{Y}$ is generically split, and hence split everywhere. To justify the last implication, each component is finite and birational over the normal integral base; it is therefore isomorphic to that base. Take a geometric point of $\tilde{T}$ lying over the generic point of T' . The corresponding fiber of $\tilde{Y}$ is a base change of the nonempty fiber of Y over that generic point. Its split cover shows

$$g \in (M\overline{P_0})^{\times \ell}. \tag{5.3}$$

This reasoning applies to each g separately, while T' , M , and P_0 stay fixed. Lemma 5.1, applied to the regular extension M/P_0 , now gives the asserted containment in V_M . $\square$

5.3 Descent to the original function field

We apply incidence to the image of a curve subfield under Θ . The last step below uses the equality $M = LP_0$ to descend all the way to L , rather than merely finding a curve field after a finite extension.

Theorem 5.4 (Correspondence of curve subfields). *For every curve subfield $E \subset K$ there is a unique curve subfield $P \subset L$ such that $\Theta(V_E) = V_P$. This assignment is a bijection between the curve subfields of K/k and those of L/l .*

Proof. Fix a normal integral projective model $X \subset \mathbb{P}_l^n$ of L/l . Let $E \subset K$ be a curve subfield, choose $t \in E \setminus k$, and put $H = \Theta(V_E)$. For $a \in k$, let $h_a = \Theta([t - a])$, and let $\mathcal{S}$ be the union of their mod- ℓ supports on X . Proposition 4.4 bounds the degrees of these supports, and hence the degree of every prime divisor in $\mathcal{S}$.

The set $\mathcal{S}$ is infinite. To see this, first observe that the kernel of all divisor-order maps on X is finite dimensional. If it contained more than $2G_X$ independent classes, apply Proposition 4.3 to representatives of a finite independent subset of that size. Their independent residues on the testing curve would all have zero point orders, contradicting Lemma 3.4 and its genus bound. On the other hand, the $[t - a]$ span an infinite-dimensional subspace of V_E , and therefore the h_a span an infinite-dimensional subspace of V_L . If $\mathcal{S}$ were finite, their order vectors would lie in a finite dimensional space; the finite-dimensional kernel just proved would give a contradiction.

We verify (5.1). For each $D \in \mathcal{S}$, some h_a has nonzero order at D . Match the valuation of D to the quasi-divisorial valuation v of K using Proposition 2.7. Its inertia I_v has nonzero restriction to E . Consequently $vE/vk \neq 0$; this group is a subgroup of the cyclic group vK/vk , and the valuation transcendence-degree inequality gives $Ev = kv$. Restriction of D_v to E therefore factors through vE/vk and has dimension at most one. Explicitly, an element whose value is in vk can be rescaled by a constant to a unit; its residue is in kv , so a further constant rescaling makes it a principal unit. Since inertia already has nonzero restriction, the images of I_v and D_v on E are the same line.

Now let $g \in L^\times$ with $[g] \in H$. Whenever $\text{ord}_D(g) = 0$, inertia at D annihilates this class. The preceding equality of restriction images, transported through Φ , implies that decomposition at D annihilates it too. By Lemma 2.4, its residue is an ℓ -th power in $l(D)$. Since a fixed rational function has nonzero integral order at only finitely many prime divisors, this proves (5.1). Lemma 5.3 now supplies M/L and $P_0 \subset M$ with $M = LP_0$, M/P_0 regular, and the image of H contained in the image of V_{P_0} .

Choose a finite normal extension N/L containing M , allowing inseparability, and let P_N be the relative algebraic closure of P_0 in N . It is a curve subfield of N/l . The image H_N of H in V_N is infinite dimensional, by Lemma 4.1, and lies in V_{P_N} . Every L -automorphism γ of N fixes H_N pointwise. Thus $H_N \subset V_{P_N} \cap V_{\gamma(P_N)}$, and Lemma 5.2 forces $\gamma(P_N) = P_N$.

Put $P_L = P_N \cap L$. This is relatively algebraically closed in L . It also has transcendence degree one over l . Indeed the invariant field $P_N^{\text{Aut}(N/L)}$ has transcendence degree one, and it is contained in the fixed field $N^{\text{Aut}(N/L)}$. For a finite normal extension the latter is L in characteristic zero and is a finite purely inseparable extension of L in positive characteristic. In the latter case, a uniform p -power of the invariant field lies in $P_N \cap L$, preserving transcendence degree. Thus P_L is a curve subfield of L .

We now use the compositum identity to obtain containment already in V_L . Since $P_L \subset L \subset M$ and every element of $P_L \subset P_N$ is algebraic over P_0 , relative algebraic closedness of P_0 in M gives $P_L \subset P_0$. The extension P_0/P_L is therefore finite algebraic. Choose the algebraic closures inside a common algebraically closed overfield. Then

$$M\overline{P_0} = L\overline{P_L}. \quad (5.4)$$

Here we used both $M = LP_0$ and the fact that P_0 and P_L have the same algebraic closure in that overfield.

The extension L/P_L is regular. Relative algebraic closedness is already known, and separability is automatic in characteristic zero. In characteristic $p > 0$, choose a separating variable u for P_L/l . It cannot become a p -th power in L : a p -th root would be algebraic over P_L and hence belong to P_L , contrary to the choice of u . Over the perfect field l , such an element extends to a separating transcendence basis of L/l , by the usual p -basis criterion. Thus $L/l(u)$, and therefore L/P_L , is separable.

For each $[g] \in H$, Lemma 5.3 and (5.4) give an ℓ -th root of g in $L\overline{P_L}$. Applying Lemma 5.1 once more, now to L/P_L , yields the exact inclusion $\Theta(V_E) \subset V_{P_L}$.

Apply the same argument to Θ^{-1} and P_L . For a curve subfield E_1 of K , it gives

$$\Theta(V_E) \subset V_{P_L} \subset \Theta(V_{E_1}).$$

Lemma 5.2 and the infinite dimension of V_E force $E = E_1$. Both inclusions are consequently equalities. The same argument with K and L reversed proves surjectivity of the matching, and Lemma 5.2 proves its uniqueness. $\square$

6 True divisors and rational subfields

The curve-subfield correspondence lets us distinguish valuations trivial on the constants from the quasi-divisorial valuations recovered in Section 2. Their restrictions to curve subfields will then identify a family of rational functions with enough point information to recover field operations.

A *true prime divisor*, or *divisorial valuation*, of F/κ is a quasi-prime divisor trivial on κ . Its value group is $\mathbb{Z}$; we use the discrete normalization when writing its order character.

The true-divisor criterion below has as a methodological antecedent Topaz's rational-character criterion [14, Lemma A.7], via the companion reconstruction manuscript [6, Sections 4–6]. Here finite coefficients require the finite-dimensional-kernel argument given below; the rational-coefficient statement is not being invoked as a proof of this proposition.

Proposition 6.1 (Recognition of true divisors). *Let F/κ be one of K/k or L/l . A quasi-divisorial valuation v of F/κ is a true prime divisor if and only if there is a curve subfield $E \subset F$ for which*

$$\dim_{\Lambda}(V_E \cap D_v^{\perp}) < \infty. \quad (6.1)$$

Consequently Φ matches the inertia and decomposition groups of true prime divisors of L/l with those of K/k .

Proof. Suppose first that v is trivial on κ . Since the residue field has transcendence degree $d - 1 \geq 1$, choose a unit t with transcendental residue, and let E be the relative algebraic closure of $\kappa(t)$ in F . The valuation is trivial on $\kappa(t)$ and therefore on its algebraic extension E . Reduction

embeds E into Fv . Its induced map $V_E \rightarrow V_{Fv}$ has finite-dimensional kernel by Lemma 4.1. Lemma 2.4 identifies that kernel with $V_E \cap D_v^\perp$, proving (6.1).

Conversely, suppose that the restriction of v to κ is nontrivial, and let E be any curve subfield. Choose $x \in E \setminus \kappa$ with $v(x) \geq 0$, replacing a nonconstant element by its inverse if necessary. There are infinitely many $c \in \kappa^\times$ with $v(c) > 0$, for example the positive powers of one such element. Each $1 + cx$ is a principal unit for v . Their classes are independent in $V_{\kappa(x)}$: up to constant factors they are $[x + c^{-1}]$, with distinct zeros on the rational curve. The map to V_E has only a finite-dimensional kernel by Lemma 4.1. These principal units therefore span an infinite-dimensional subspace of $V_E \cap D_v^\perp$. This disproves (6.1) for every E . The final assertion now follows from Theorem 5.4 and the already established matching of quasi-divisorial pairs. $\square$

For a curve subfield $E \subset F$, restriction of characters gives a continuous surjection $A_F \rightarrow A_E$, dual to the inclusion $V_E \subset V_F$. If a true divisor restricts nontrivially as a valuation on E , that restriction is a positive integer multiple of a point valuation of the smooth projective curve of E . Its inertia image in A_E is either the corresponding point line or zero; the latter occurs when that integer is divisible by ℓ . Thus we retain the nonzero inertia images when discussing which point lines are detected.

Definition 6.2. A curve subfield $E \subset F$ is *good* if it is rational over the constants and every point line in A_E is the nonzero image of the inertia of some true divisor of F . A *good function* is an element $t \in F$ generating a good subfield $\kappa(t)$.

Proposition 6.3 (Recognition of good subfields). *The curve-subfield correspondence of Theorem 5.4 restricts to a bijection on good subfields. For matched good subfields $E \subset K$ and $P \subset L$, the induced character isomorphism gives a bijection between the points of their smooth projective rational curves.*

Proof. For each curve subfield E , let $\mathcal{Q}_E$ be the collection of nonzero images of true divisorial inertia in A_E . By Proposition 6.1, these collections correspond under the induced character isomorphisms. They are subfamilies of the point lines of E . Choose a nonzero generator of each line. The generators tend to zero outside finite sets, so summation defines a continuous map from a product of copies of Λ into A_E .

By Lemma 3.4, when every point occurs, this summation map has one-dimensional kernel, generated by a coefficient family with every coordinate nonzero. A proper subfamily of the point lines has no relation: extending any proposed relation by zero to the missing points would contradict the description of the full relation kernel. Hence the stated kernel property recognizes exactly when $\mathcal{Q}_E$ contains every point line. Once this holds, the common kernel of their orders on V_E has dimension $2g(E)$. Its vanishing recognizes genus zero, equivalently rationality over the algebraically closed constants. Both conditions are preserved by the character isomorphisms. Finally, distinct points have distinct inertia lines, so their matching gives the asserted point bijection. $\square$

We now construct enough good functions for the later reconstruction. The construction is local on a smooth model: a ratio of two transverse parameters has a multiplicity-one divisor over every value, and the exceptional divisor of the blowup ensures that the resulting rational subfield is relatively algebraically closed.

Lemma 6.4 (A supply of good functions). *Let F/κ be a finitely generated extension of an algebraically closed field with $\text{trdeg}(F/\kappa) \geq 2$ and $\text{char}(\kappa) \neq \ell$. On an integral model of F/κ , let*

q be a smooth closed point and let a, b be rational functions regular at q , vanishing there, with independent differentials in the cotangent space at q . Then $t = a/b$ is a good function. Moreover, for every $f \in F^\times$ there is a good function t such that ft is good. Consequently good functions generate F over κ , and their classes span V_F .

Proof. Blow up the smooth point q and let v be the valuation of the exceptional divisor. Write $e = \text{trdeg}(F/\kappa)$. Its residue field is the rational field of $\mathbb{P}_\kappa^{e-1}$. The leading linear forms of a and b give independent homogeneous coordinates on the exceptional divisor. Their ratio is the residue of t and is a rational coordinate in Fv . In particular v is trivial on $\kappa(t)$, and $\kappa(\bar{t})$ is relatively algebraically closed in Fv . The relative algebraic closure of $\kappa(t)$ in F is a finite algebraic extension on which v is still trivial. It therefore embeds by reduction into Fv , over $\kappa(t) \cong \kappa(\bar{t})$. Relative algebraic closedness in the residue field forces this extension to be $\kappa(t)$ itself.

For each $c \in \kappa$, the function $a - cb$ has nonzero linear term at q and cuts out a prime divisor locally there, with order one. The function b is a unit at its generic point because the linear terms of a and b are independent. The resulting true divisor has $\text{ord}(t - c) = 1$ and detects the point c of $\mathbb{P}_\kappa^1$. The divisor locally cut out by b similarly detects infinity: a is a unit at its generic point and t has a simple pole. Every point line is thus detected, proving that t is good.

Given $f \in F^\times$, choose the smooth point q in an open set where f is a unit, and choose a, b as above. The functions fa, b also vanish at q with independent differentials, since $d(fa)_q = f(q) da_q$. Hence both $ft = (fa)/b$ and $t = a/b$ are good. The equality $f = (ft)/t$ proves field generation, and $[f] = [ft] - [t]$ proves the spanning assertion for V_F . $\square$

Lemma 6.5 (A common partner). *For any finite list of good functions $t_1, \dots, t_m$ of F/κ there is a good function s such that s is algebraically independent of each t_i over κ and*

$$t_i + cs \text{ is good for every } i \text{ and every } c \in \kappa^\times.$$

Proof. Choose a smooth closed point q on a model where all the t_i are regular, and choose a, b vanishing at q with independent differentials. Put $s = a/b$. It is good by Lemma 6.4. For $c \neq 0$, write

$$t_i + cs = \frac{t_i b + ca}{b}.$$

At q the numerator has differential $t_i(q) db_q + c da_q$, which is independent of db_q . The same lemma therefore makes every indicated sum good.

Let v be the exceptional-divisor valuation at q used in the preceding lemma. The residue of s is transcendental over κ . By contrast, the restriction of v to $\kappa(t_i)$ is centered at the finite point $t_i(q)$: the nonzero function $t_i - t_i(q)$ has positive order at q . Its residue field is consequently κ . If s were algebraic over $\kappa(t_i)$, relative algebraic closedness of that good subfield in F would imply $s \in \kappa(t_i)$, contradicting its transcendental residue. This proves the required independence. $\square$

7 Recovering the field from simultaneous values

The preceding section recovers the good rational subfields and the points of their projective lines. We now recover the field operations from this information. A single true divisor can detect the values of several good functions at once. The resulting correspondence of algebraic relations will first identify the constant fields and then give an isomorphism of the perfect closures.

For every good function $t \in K$, choose a generator $t' \in L$ of the good rational subfield matched with $k(t)$ by Proposition 6.3. Its point correspondence, expressed in these coordinates, is a bijection

$$B_t : \mathbb{P}^1(k) \longrightarrow \mathbb{P}^1(l).$$

Changing t' by a projective linear transformation, we arrange $B_t(\infty) = \infty$. These choices are made independently for the different good functions. At this stage the maps B_t are only bijections of sets.

7.1 Simultaneous values

For a finite tuple $\mathbf{t} = (t_1, \dots, t_m)$ of good functions, let $X_{\mathbf{t}} \subset (\mathbb{P}_k^1)^m$ be the reduced closure of the image of the rational map defined by the t_i on a model of K/k . Thus $X_{\mathbf{t}}$ is the irreducible variety of algebraic relations among the t_i . Define $Y_{\mathbf{t}} \subset (\mathbb{P}_l^1)^m$ in the same way from the chosen t'_i .

Lemma 7.1 (Simultaneous values). *For every finite tuple $\mathbf{t}$ of good functions there are dense open subsets $U \subset X_{\mathbf{t}}$ and $U' \subset Y_{\mathbf{t}}$ such that*

$$\left(\prod_i B_{t_i} \right) (U(k)) \subset Y_{\mathbf{t}}(l), \quad \left(\prod_i B_{t_i}^{-1} \right) (U'(l)) \subset X_{\mathbf{t}}(k). \quad (7.1)$$

Proof. Choose a smooth model open on which all the t_i are regular and all the differentials dt_i are nowhere zero. Such an open exists. In positive characteristic, t_i cannot be a p -th power in K , since its p -th root would be algebraic over the relatively algebraically closed subfield $k(t_i)$. As k is perfect, this gives $dt_i \neq 0$; in characteristic zero the same conclusion is immediate.

At a closed point P of this open, write $a_i = t_i(P)$. The exceptional divisor of the blowup at P defines a true divisorial valuation v with

$$\text{ord}_v(t_i - a_i) = 1 \quad \text{for every } i.$$

Indeed $dt_i(P) \neq 0$ says that $t_i - a_i$ has order one in the maximal ideal of the regular local ring at P . Thus restriction of I_v to each $k(t_i)$ is the nonzero point line at a_i .

By Proposition 6.1, the matched valuation on L is a true divisor. Restriction of characters commutes with the correspondence of good subfields, so its center in the t'_i -line is $B_{t_i}(a_i)$ for every i . Specializing the target tuple along this valuation shows that $(B_{t_i}(a_i))_i$ belongs to $Y_{\mathbf{t}}(l)$: the generic tuple lies in that closed subvariety of the product of projective lines, and so does its specialization.

The image of the chosen model open in $X_{\mathbf{t}}$ is constructible and dense, hence contains a dense open subset U . Every k -point of U has a closed preimage, since its nonempty fiber is of finite type over the algebraically closed field k . This proves the first inclusion in (7.1). The same argument for Θ^{-1} proves the second. $\square$

These inclusions provide information in the Zariski topology without asserting that any B_t preserves that topology. We next apply them to the relation $z = t + cs$. This will force the coordinate bijections to have a common field-theoretic form.

7.2 Aligning the constant fields

An element of $l(X_1, \dots, X_r)^i$ will be called a *perfect rational function*. It has a well-defined value at a general tuple of l -points: in positive characteristic one first raises it to a sufficiently large p -power and then takes the unique corresponding root in l . Here and below, *general* means belonging to a suitable dense open.

Let $\mathcal{H}_l$ be the group of permutations of $\mathbb{P}^1(l)$ generated by $\mathrm{PGL}_2(l)$, together with $x \mapsto x^p$ when $\mathrm{char} l = p > 0$.

Lemma 7.2. *A perfect rational function in one variable that is injective on a cofinite subset of $\mathbb{P}^1(l)$ agrees there with a unique member of $\mathcal{H}_l$. In characteristic zero, $\mathcal{H}_l = \mathrm{PGL}_2(l)$; in characteristic $p > 0$, every member has a unique expression*

$$M \circ \mathrm{Frob}^n, \quad M \in \mathrm{PGL}_2(l), \quad n \in \mathbb{Z}.$$

In particular, $\mathcal{H}_l$ embeds in the group of permutations of $\mathbb{P}^1(l)$ modulo agreement outside finite sets.

Proof. In characteristic zero, injectivity on a cofinite set implies that a rational map has degree one. In characteristic p , choose $N \geq 0$ such that the given function h satisfies $h^{p^N} = r \in l(X)$. Write $r = q^{p^a}$, with $a \geq 0$ maximal and $q \in l(X)$ separating. The map q is also injective on a cofinite set, so its degree is one. Indeed a separating map of degree greater than one has more than one distinct point over a general value, and deleting finitely many points cannot change this. Thus q is projective linear, and $h = \mathrm{Frob}^{a-N} \circ q$ on points.

Conjugating a projective linear transformation by Frobenius raises its coefficients to p -th powers. This proves the asserted normal form. No nonzero power of Frobenius is projective linear: for a positive exponent the corresponding map has inseparable degree greater than one, and a negative exponent reduces to this case by inversion. The normal form is therefore unique. Finally, two perfect rational functions agreeing cofinitely are equal, as is seen after clearing Frobenius powers and comparing ordinary rational functions. $\square$

Write $T_a(x) = x + a$ and $m_c(x) = cx$, fixing ∞ ; these transformations generate $\mathrm{Aff}(k)$. We compare bijections of projective point sets modulo finite disagreement. Composition is well defined with this convention, because bijections carry finite sets to finite sets. Membership in $\mathcal{H}_l$ in the next lemma means membership after this identification. Lemma 7.2 ensures that the representing member of $\mathcal{H}_l$ is unique.

Lemma 7.3 (Transported addition). *Let t, s be algebraically independent good functions such that $z_c = t + cs$ is good for every $c \in k^\times$. For every $c \in k^\times$ there is a perfect rational function $R_c \in l(X, Y)^i$ such that*

$$R_c(x, y) = B_{z_c}(B_t^{-1}(x) + cB_s^{-1}(y)) \quad \text{for general } (x, y) \in l^2. \quad (7.2)$$

The one-variable slices of these functions give

$$B_s B_t^{-1} \in \mathcal{H}_l, \quad B_t \mathrm{Aff}(k) B_t^{-1} \subset \mathcal{H}_l. \quad (7.3)$$

The second inclusion is an embedding of groups.

Proof. Consider the relation variety Y for (t, s, z_c) . Its projection to the first two factors is dominant. Indeed $k(t)$ and $k(s)$ are distinct curve subfields, so their matched subfields $l(t')$ and $l(s')$ are distinct. Any two distinct curve subfields have compositum of transcendence degree two: if their compositum had transcendence degree one, relative algebraic closedness would make them equal.

On the dense open of Y given by the inverse inclusion in Lemma 7.1, the third coordinate is forced by the first two to be

$$B_{z_c}(B_t^{-1}(x) + cB_s^{-1}(y)).$$

We restrict to finite coordinates, which is permitted because all the coordinate functions are nonconstant and the B 's fix ∞ . If $\dim Y = 3$, then $Y = (\mathbb{P}_l^1)^3$, and a dense open has infinitely many third coordinates above a general pair (x, y) , a contradiction. Consequently $\dim Y = 2$.

The projection $Y \rightarrow (\mathbb{P}_l^1)^2$ is proper, dominant and generically finite. The complement of the open just used has dimension at most one, so its image is a proper closed subset of the base. After removing that image and restricting to the locus of finite fibers, every closed fiber has precisely one point. Thus the finite extension $l(Y)/l(x, y)$ has separable degree one. Indeed, in positive characteristic write the minimal polynomial of the third coordinate as $g(Z^{p^e})$, with g separable; in characteristic zero take the minimal polynomial itself. After shrinking the base to preserve the degree and nonzero discriminant of the separable polynomial, the separable degree counts the distinct points of a general fiber, since taking p^e -th roots is a bijection on algebraically closed points. The extension is therefore purely inseparable, and the third coordinate gives a perfect rational function R_c satisfying (7.2).

Fixing a general y in (7.2) gives a perfect rational function of x , agreeing cofinitely with a bijection. A dense open in $(\mathbb{P}_l^1)^2$ has cofinite fibers outside finitely many values of y . Since B_s is a bijection, Lemma 7.2 therefore gives

$$D_b := B_{z_c}T_{cb}B_t^{-1} \in \mathcal{H}_l \quad \text{for cofinitely many } b \in k.$$

For two such parameters,

$$D_b^{-1}D_{b'} = B_tT_{c(b'-b)}B_t^{-1}.$$

Every element of k is a difference of two members of a cofinite subset: for any prescribed difference, the two required cofinite sets intersect. Hence all translations conjugated by B_t belong to $\mathcal{H}_l$. The products $D_bD_{b'}^{-1}$ give the same assertion for B_{z_c} , and then D_b gives $B_{z_c}B_t^{-1} \in \mathcal{H}_l$.

Fixing a general x instead yields

$$B_{z_c}T_a m_c B_s^{-1} \in \mathcal{H}_l \quad \text{for cofinitely many } a \in k.$$

Composing on the left successively with

$$B_{z_c}T_{-a}B_{z_c}^{-1} \quad \text{and} \quad B_tB_{z_c}^{-1},$$

both already in $\mathcal{H}_l$, gives

$$B_t m_c B_s^{-1} \in \mathcal{H}_l \quad (c \neq 0).$$

At $c = 1$ this gives $B_s B_t^{-1} \in \mathcal{H}_l$. Composing again gives all conjugated multiplications $B_t m_c B_t^{-1}$. This proves (7.3). Finally, two distinct affine transformations cannot agree outside a finite set; conjugating by a bijection preserves this property. Thus the resulting homomorphism is injective. $\square$

We have obtained an affine group action by perfect rational transformations. Its translation subgroup will identify the addition on the constants. The two-variable function R_1 is then needed to show that the resulting embedding of constant fields is onto.

Proposition 7.4 (Alignment of the constants). *The fields k and l have the same characteristic. There is a field isomorphism $\sigma : k \rightarrow l$ such that, for every good function t , there is $H_t \in \mathcal{H}_l$ with*

$$B_t = H_t \circ \sigma \quad \text{outside a finite subset of } \mathbb{P}^1(k), \quad (7.4)$$

where $\sigma(\infty) = \infty$.

Proof. Choose a good function t , and use Lemma 6.5 to choose a good s satisfying the hypotheses of Lemma 7.3. First consider the embedding $B_t \text{Aff}(k) B_t^{-1} \subset \mathcal{H}_l$, with the finite-agreement convention of that lemma.

The affine image is projective linear. In positive target characteristic the Frobenius exponent defines a homomorphism $\mathcal{H}_l \rightarrow \mathbb{Z}$. Its restriction to the affine image is zero: $k^\times$ is divisible, while the additive group of k is divisible in characteristic zero and torsion in positive characteristic. Both groups therefore have zero image in $\mathbb{Z}$, and they generate $\text{Aff}(k)$. In characteristic zero for l , there is no exponent to consider. We have in either case an embedding into $\text{PGL}_2(l)$.

Translations yield a field embedding. Let U be the image of the translation subgroup. This is an infinite abelian subgroup of $\text{PGL}_2(l)$, normalized by the whole affine image. It cannot contain a nonidentity semisimple element. To see this, such an element has two fixed points, and every element of U preserves their pair. The subgroup fixing both points has index at most two in U ; it is infinite, hence contains an element of order greater than two. The centralizer of that element is the torus fixing the two points, so all of U lies in that torus. Any transformation normalizing U preserves the same pair and acts on U by either identity or inversion. This contradicts the faithful conjugation action of $k^\times$ on its additive group in $\text{Aff}(k)$.

Every nonidentity element of U is consequently unipotent. Conjugate the image in $\text{PGL}_2(l)$ so that one such element is a translation with unique fixed point ∞ . Its centralizer consists of translations, and hence so does U . Write this conjugation as postcomposition $B = M B_t$, with $M \in \text{PGL}_2(l)$. We obtain an injective additive map $\chi : k \rightarrow l$ for which

$$B T_b B^{-1} = T_{\chi(b)} \quad \text{modulo finite disagreement.} \quad (7.5)$$

After rescaling the target coordinate, assume $\chi(1) = 1$. The image of m_c normalizes U , hence fixes ∞ , and is an affine transformation of some slope λ_c . Conjugating translations gives

$$\chi(cb) = \lambda_c \chi(b).$$

Setting $b = 1$ gives $\lambda_c = \chi(c)$. Thus χ is multiplicative as well as additive and is a field embedding. In particular, the characteristics of k and l agree.

The field embedding is onto. Set $z = t + s$. Lemma 7.3 gives members $M_s, M_z \in \mathcal{H}_l$ such that

$$B_s = M_s B, \quad B_z = M_z B \quad \text{modulo finite disagreement.}$$

Using R_1 from (7.2), define the perfect rational function

$$F(X, Y) = M_z^{-1}(R_1(M^{-1}X, M_s Y)).$$

For cofinitely many $b \in k$, and for each such b for cofinitely many $X \in l$, equations (7.2) and (7.5) give

$$F(X, B(b)) = B(B^{-1}(X) + b) = X + \chi(b). \quad (7.6)$$

Here is the order of the exclusions. First exclude the finitely many values of b at which $B(b)$ is infinite, the equality $B_s = M_s B$ fails, or the corresponding vertical fiber misses the open of (7.2). For each remaining b , that open excludes only finitely many X ; the equality $B_z = M_z B$ excludes finitely many more, since its argument is a bijective function of X . Finally (7.5) excludes a finite set of X , which may depend on b .

For each fixed remaining b , (7.6) is therefore an identity of perfect rational functions in X . Choose $X_0 \in l$ such that $F(X_0, Y)$ is defined and finite for general Y . This choice is possible by expressing a power of F as a quotient of polynomials and choosing X_0 for which its denominator does not vanish identically as a polynomial in Y . Define

$$h(Y) = F(X_0, Y) - X_0.$$

For all but finitely many of the remaining b , the denominator is nonzero at $Y = B(b)$; evaluating the rational identity then gives

$$\chi(b) = h(B(b)).$$

In particular, this evaluation does not require X_0 to avoid the original exceptional sets for all b at once.

The injectivity of χ implies that h is injective on a cofinite subset of $\mathbb{P}^1(l)$. By Lemma 7.2, it represents a member of $\mathcal{H}_l$, so $\chi(k)$ contains a cofinite subset of l . An infinite proper subfield cannot be cofinite: if $a \notin \chi(k)$, the infinite coset $a + \chi(k)$ is disjoint from it. Thus $\chi(k) = l$. Taking $\sigma = \chi$, the equality $\sigma = hB = hMB_t$ outside a finite set proves (7.4) for our initial t .

The same constant isomorphism works for every good function. For any other good function r , Lemma 6.5 supplies one good s_0 that is algebraically independent of both t and r , with $t + cs_0$ and $r + cs_0$ good for every $c \neq 0$. Lemma 7.3 gives $B_{s_0} B_t^{-1}, B_{s_0} B_r^{-1} \in \mathcal{H}_l$, hence $B_r B_t^{-1} \in \mathcal{H}_l$. The alignment already obtained for t therefore gives (7.4) for r , with the same σ . $\square$

7.3 The isomorphism of perfect closures

Fix σ and the H_t given by Proposition 7.4, and define

$$u_t = H_t^{-1}(t') \in L^i \quad \text{for every good function } t.$$

These are field elements, obtained by projective linear operations and Frobenius powers. They satisfy

$$l(u_t)^i = l(t')^i. \tag{7.7}$$

We show that the assignments $t \mapsto u_t$ preserve every algebraic relation. This is the point where the finite exceptions in the individual coordinate alignments cease to matter.

Theorem 7.5. *There is a field isomorphism*

$$\alpha : K^i \xrightarrow{\sim} L^i, \quad \alpha|_k = \sigma,$$

such that $\alpha(t) = u_t$ for every good function t . In particular, if $k(t)$ and $l(t')$ are matched good rational subfields, then $\alpha(k(t)^i) = l(t')^i$.

Proof. Let $\mathbf{t} = (t_1, \dots, t_m)$ be a finite tuple of good functions, and let $Z_{\mathbf{t}} \subset (\mathbb{P}_l^1)^m$ be the relation variety of $(u_{t_1}, \dots, u_{t_m})$. We first prove

$$\sigma(X_{\mathbf{t}}(k)) = Z_{\mathbf{t}}(l), \tag{7.8}$$

where σ acts coordinatewise.

The point map

$$\mathcal{H} = \prod_i H_{t_i}^{-1} : (\mathbb{P}^1(l))^m \longrightarrow (\mathbb{P}^1(l))^m$$

is a Zariski homeomorphism. Projective linear maps are isomorphisms; in positive characteristic the coordinate map $[X : Y] \mapsto [X^p : Y^p]$ is finite, radicial and surjective, hence a homeomorphism on point spaces, as is its inverse. Products with different powers in different coordinates have the same property. It follows that

$$\mathcal{H}(Y_{\mathbf{t}}(l)) = Z_{\mathbf{t}}(l).$$

To justify this last equality at the level of relation varieties, place the finite tuple of u_{t_i} in one finite purely inseparable extension of L . On a common model, the tuples $(t'_i)_i$ and $(u_{t_i})_i$ are related by $\mathcal{H}$ wherever they are defined. Their images are dense in their respective relation varieties. Taking closures under the homeomorphism gives the equality.

Apply now the first inclusion of Lemma 7.1. Delete from its source open the finitely many exceptional values in each coordinate where (7.4) fails. Each deleted coordinate fiber is a proper closed subset, since every t_i is nonconstant. The resulting open is still dense, and on it $\mathcal{H} \circ \prod_i B_{t_i} = \sigma$. Consequently a dense subset of $\sigma(X_{\mathbf{t}}(k))$ lies in $Z_{\mathbf{t}}(l)$. The former is the point set of the variety obtained from $X_{\mathbf{t}}$ by applying σ to coefficients, so taking closures gives

$$\sigma(X_{\mathbf{t}}(k)) \subset Z_{\mathbf{t}}(l).$$

For the reverse inclusion, use the second inclusion of Lemma 7.1. Delete the finitely many target coordinate values at which $B_{t_i}^{-1} = \sigma^{-1}H_{t_i}^{-1}$ fails. Again this leaves a dense open, now in $Y_{\mathbf{t}}$, because each target coordinate is nonconstant. Its image under $\mathcal{H}$ is dense in $Z_{\mathbf{t}}$ and lies in $\sigma(X_{\mathbf{t}}(k))$. Taking closures proves (7.8).

Thus, for every polynomial $P \in k[T_1, \dots, T_m]$,

$$P(t_1, \dots, t_m) = 0 \iff \sigma(P)(u_{t_1}, \dots, u_{t_m}) = 0,$$

where $\sigma(P)$ is obtained by applying σ to the coefficients. Good functions generate K by Lemma 6.4. The displayed equivalence for every finite tuple makes the assignments $a \mapsto \sigma(a)$ for $a \in k$ and $t \mapsto u_t$ for good t well defined on all polynomial expressions, and preserves their nonzero values. Passing to quotients therefore gives an embedding $\alpha_0 : K \hookrightarrow L^i$.

Because L^i is perfect, this embedding extends uniquely to an embedding $\alpha : K^i \hookrightarrow L^i$. Its image contains $l = \sigma(k)$ and every u_t , and is itself perfect; by (7.7) it contains the perfection of every matched good rational subfield of L . All good rational subfields are matched, and their generators generate L . The image of α therefore contains L^i , proving surjectivity and the final assertion. $\square$

8 Compatibility and uniqueness

Theorem 7.5 gives an isomorphism of perfect fields that carries the perfection of every good rational subfield to the perfection of its match under Θ . We must prove that its action on all multiplicative classes is Θ up to a single scalar. We then determine which field automorphisms act by scalars.

8.1 Distinguishing divisors by rational subfields

For a true prime divisor v and a good rational subfield $E \subset K$, consider the restriction $D_v \rightarrow A_E$. Its rank here means the dimension of its image as a Λ -vector space.

Lemma 8.1. *Let v be a true prime divisor of K/k and let $E \subset K$ be a curve subfield. The restriction $v|_E$ is trivial if and only if the image of $D_v \rightarrow A_E$ is infinite-dimensional. If $v|_E$ is nontrivial, this image has dimension at most one.*

Proof. If $v|_E$ is trivial, reduction embeds E in Kv . The kernel of $V_E \rightarrow V_{Kv}$ is finite-dimensional by Lemma 4.1, whereas V_E is infinite-dimensional. Dualizing and using Lemma 2.4 shows that the restriction of D_v to A_E has infinite-dimensional image.

Otherwise $v|_E$ is a positive multiple of the order at a point x of the smooth projective curve of E/k . If f is a unit there, its residue is some $c \in k^\times$, and f/c is a principal v -unit in K . Every character in D_v therefore kills f . Its restriction to E factors through the value group of $v|_E$, which is cyclic. The image has dimension at most one. This argument uses decomposition characters; it remains valid if the ramification multiplicity is divisible by ℓ . $\square$

Lemma 8.2. *For distinct true prime divisors v_1, v_2 of K/k , there is a good rational subfield $E \subset K$ such that $v_1|_E$ is trivial and $v_2|_E$ is nontrivial.*

Proof. Choose a normal projective model X on which both valuations have codimension-one centers D_1, D_2 . We recall why this is possible. For each valuation, lifts of a residue transcendence basis, together with a uniformizer, define a rational subfield over which K is finite. Normalization of a suitable projective model of that subfield realizes the valuation as a prime divisor. A common normal projective model dominating the two models retains codimension-one centers: the residue field of each new center contains that of the old center, of transcendence degree $d - 1$ over k .

Choose two distinct closed points P, Q of $D_1 \setminus D_2$, and a smooth closed point R of $X \setminus (D_1 \cup D_2)$. The first choice is possible because $d \geq 2$. There is an affine open containing these points and meeting D_2 ; write its coordinate ring as A , and let $\mathfrak{p}_2$ be the prime ideal of D_2 there. Choose two independent cotangent vectors at R . The Chinese remainder theorem, applied to $\mathfrak{p}_2, \mathfrak{m}_P, \mathfrak{m}_Q, \mathfrak{m}_R^2$, produces $a, b \in A$ satisfying

	D_2	P	Q
a	0	0	1
b	1	1	1

and having value zero and the prescribed independent first-order terms at R . These ideals are pairwise comaximal, so the prescriptions are independent. By Lemma 6.4, $t = a/b$ is good. Its residue on D_1 is nonconstant, since it takes different values at P, Q , and its value along D_2 is positive. Thus v_1 is trivial on $k(t)$ and v_2 is nontrivial there. $\square$

8.2 The scalar is global

Proposition 8.3. *Let $\Psi : V_K \rightarrow V_K$ be a compatible Λ -linear automorphism. If $\Psi(V_E) = V_E$ for every good rational subfield $E \subset K$, then $\Psi = a \text{ id}$ for one $a \in \Lambda^\times$.*

Proof. By Proposition 6.1, Ψ^* permutes the pairs $I_v \subset D_v$ for true prime divisors. Since Ψ fixes each V_E , it preserves the rank of restriction of D_v to A_E . Lemmas 8.1 and 8.2 therefore force this permutation of true divisors to be the identity.

Fix a good rational subfield E . Every point line of A_E is the nonzero restriction of some I_v . The induced dual automorphism $\Psi_E^* = (\Psi|_{V_E})^*$ of A_E therefore fixes every point line. Write

$$\Psi_E^*(\text{ord}_x) = a_x \text{ord}_x, \quad a_x \in \Lambda^\times, \quad x \in C_E(k),$$

where C_E denotes the smooth projective model of E . The sum of these order characters is zero, and its coefficient families have exactly the diagonal kernel by Lemma 3.4. Continuity permits applying Ψ_E^* to this sum, so all a_x equal one scalar a_E . Since E is rational, its point orders separate V_E , and hence $\Psi|_{V_E} = a_E \text{id}$.

Let $E = k(t)$ and $F = k(s)$ be any two good rational subfields. On a common smooth model open, t, s are regular and both differentials are nonzero. Blow up a closed point of that open. Its exceptional divisor v has

$$\text{ord}_v(t - t(P)) = \text{ord}_v(s - s(P)) = 1.$$

Thus the inertia line I_v restricts nontrivially to a point line of both A_E and A_F . The scalar by which Ψ^* acts on I_v must be both a_E and a_F . All these scalars agree. Good-function classes span V_K by Lemma 6.4, proving the claim. $\square$

Apply the proposition to $\Psi = \alpha_1^{-1}\Theta$, where $\alpha : K^i \rightarrow L^i$ is supplied by Theorem 7.5. The canonical identifications under perfection are compatible with Milnor multiplication. Also $\alpha(E^i)$ is the perfection of the good field matched to E , so Ψ fixes V_E for every such E . Consequently $\Theta = a\alpha_1$ for a single $a \in \Lambda^\times$.

8.3 Uniqueness up to Frobenius

Proposition 8.4. *Suppose $\beta : K^i \rightarrow K^i$ is a field automorphism preserving k whose action on V_K is a scalar. Then β is the identity in characteristic zero and an integral power of Frobenius in positive characteristic.*

Proof. Let $E = k(t)$ be good. Its perfection is relatively algebraically closed in K^i : if an element is algebraic over E^i , clearing Frobenius powers puts an algebraic element over E in K , and relative algebraic closedness then puts it in E . The same applies to $\beta(E^i)$. Choose a nonconstant element of $\beta(E^i) \cap K$ by clearing powers, and let P be the relative algebraic closure of the rational field it generates in K . Relative algebraic closedness of $\beta(E^i)$ gives $P^i \subset \beta(E^i)$. Conversely, every element of $\beta(E^i)$ is algebraic over this rational field; clearing powers puts it in P . Hence $\beta(E^i) = P^i$. Since β_1 is scalar, $V_P = V_E$ inside V_K . Lemma 5.2 forces $P = E$.

It follows that $\beta(t) = h(t)$ for a perfect rational function h generating $k(t)^i$ over k up to perfection. This implies that h belongs to the group $\mathcal{H}_k$ of projective transformations and Frobenius powers from Section 7. Indeed, after clearing powers, its separable degree must be one; a larger separable degree could not disappear on passing to perfections. In characteristic zero this says simply that h is projective linear.

Write $\rho = \beta|_k$. For any two distinct points $a, b \in \mathbb{P}^1(k)$, choose a rational function $f_{a,b}(t)$ with divisor $[a] - [b]$. The point-order vector of $\beta_1([f_{a,b}])$ is supported exactly at

$$h^{-1}(\rho(a)), \quad h^{-1}(\rho(b)).$$

Here ρ fixes infinity. For negative Frobenius powers, the order vector is interpreted by clearing powers; multiplication by a power of p is invertible in Λ . Both displayed coordinates therefore have nonzero orders. Scalar action on V_E implies that the permutation $h^{-1}\rho$ fixes every unordered

pair of distinct points. On a set with at least three points this forces every point to be fixed: intersect the fixed pairs $\{a, b\}$ and $\{a, c\}$. Thus h and ρ agree on all points.

In particular h fixes $0, 1, \infty$. An element of $\mathcal{H}_k$ with these three fixed points is a pure Frobenius power, or the identity in characteristic zero. In positive characteristic its exponent is determined by ρ and therefore is the same for every good t . Distinct Frobenius powers act differently on the infinite algebraically closed field k : equality of two powers would force every element to satisfy $X^{p^n} - X = 0$ for some $n > 0$. Good functions generate K , so β is this same Frobenius power on K and then on its perfection. $\square$

Proof of Theorem 1.1. A compatible Θ gives equal transcendence degrees by Proposition 2.6. Theorem 7.5 produces a field isomorphism $\alpha : K^i \rightarrow L^i$ respecting constants; in particular, the characteristics agree. Proposition 8.3 shows that Θ and α_1 differ by a single scalar, proving surjectivity of the stated map. If two field isomorphisms have the same image modulo scalars, their quotient satisfies Proposition 8.4, proving injectivity modulo Frobenius. Conversely, every integral Frobenius power acts on V_L by the corresponding nonzero scalar. The map is therefore well-defined and bijective with exactly the stated equivalence relations. $\square$

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